

# **ME 3043 Production and Operations Management**

## **Semester 05**

### **Final Report – G7**

**By**

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## **Executive Summary**

As the case study for the class project of the module, our group selected “Pandula Enterprises” which is a convenience store situated in Ratmalana area. We interviewed its owner and the other two employees and collected the relevant data which gave the details of their customer base and operational capabilities.

From our visits to the premises, we analysed the current operation procedure of the institution and sorted out the short comings and flaws in the existing system. With our learned techniques of production and operations management, we suggested some improvements to the facility such as rearranging the layout of the facility such that waiting time for majority of the customers to be minimum. The logic we used was simulated using Tecnomatix software and the efficiency of the existing and proposed systems were compared.

Also, we suggested a new plan for ordering stock and a warehouse for maintaining the stocks which is currently not there. Further, there are few suggestions for maintaining the quality and improving the customer attraction.

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# 1 Introduction

## 1.1 Background

Pandula enterprises is a small-scale business which initially started as a family run grocery store based on Ratmalana area in Colombo district. Currently it has expanded to a convenience store with 3 employees including the owner. The main target customers are the neighbourhood homes and factory employees of the nearby factories. The convenience store sells number of items and specific locations are arranged for them in the premises. The list of items and specific locations can be illustrated as follows.

|                           |                                                                                                                                                                                                                                                                      |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Daily needs and cosmetics | Perfumes, Hair gel, Wax, Toothpaste, Toothbrushes, Razors                                                                                                                                                                                                            |
| Fancy Items               | Snow sprays, Birthday celebration items (Candles)                                                                                                                                                                                                                    |
| Food Items                | Ice creams, Chicken & Meat products (Sausages, Meat balls, Drumsticks), Swiss rolls, Cakes, Biscuits, Canned fish, Noodles, Cooking oils, Chili powder and other culinary, Potatoes, Big onion, Garlic, Pasta, Colorings, Icing sugar, Bites, Milk and milk products |
| Stationary Items          | Books, Paper cutters, Scissors etc.                                                                                                                                                                                                                                  |
| Electric items            | Bulbs, Batteries, Insulation tape                                                                                                                                                                                                                                    |
| Cleaning Items            | Toilet Cleaners, Dish washing liquids, Soaps, Shampoo                                                                                                                                                                                                                |
| Other items               | Super glue, Incense sticks, Pahan Thira (wicks), Lunch sheets, Slippers, Masks, Brooms, Idal, Bulath wita , Newspapers & Magazines, Reloads and Recharge service                                                                                                     |
| Sports items              | Tennis Balls                                                                                                                                                                                                                                                         |
| Pharmaceuticals           | Siddhalepa, Iodex balms, Samahan, Panadol                                                                                                                                                                                                                            |
| Beverage items            | Soft drinks, Bottled water, Energy drinks (Kist Ride)                                                                                                                                                                                                                |

*Table 1: List of items in the convenience store*

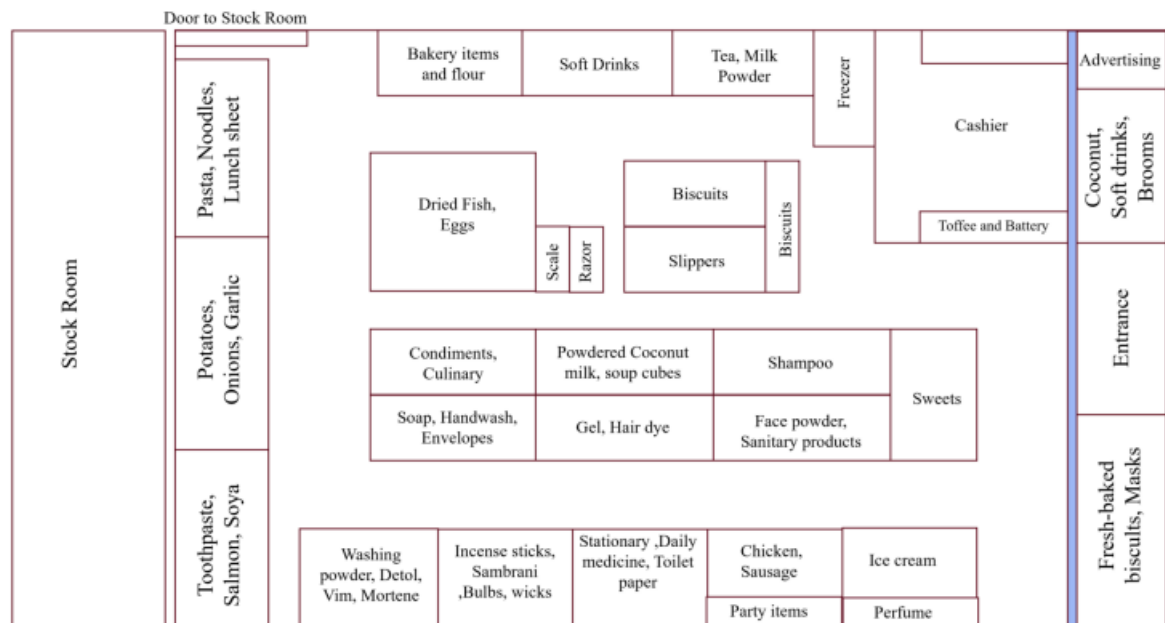


Figure 1: Existing layout of the convenience store

## 1.2 Problem Definition

After our visit to the facility, we noticed the following problems that causes the deficiency of profit growth in the store and poor stock management. [1]

- The lack of an organized shelf system.
- Inefficient utilization of the designated storage area.
- Unavailability of a proper billing system and an inventory management system.
- Inability to forecast the demands and monitor the profit due to the lack of proper documentation.

## 1.3 Project Objective and Scope

The aim of this project is to increase the productivity of the convenience store by providing solutions to the existing management problems using the theories learned during the production and operations management module.

The main objectives of this project can be listed down as follows,

1. Maximize the profit by forecasting the demand for items and ordering the optimum amount of units required.
2. Re arranging the layout of the convenience store to improve the quality of customer experience and their convenience.
3. Introduction of a proper inventory management system to conveniently monitor the stock availability.

## 2 Methodology

The process of achieving each of the above-mentioned objectives needs a systematical approach.

### 2.1 Forecasting the Demand [2]

For the forecasting to be done, the data of the sales should be collected and calculations should be done based on those data. After having several conversations with the store owner, the required data was collected.

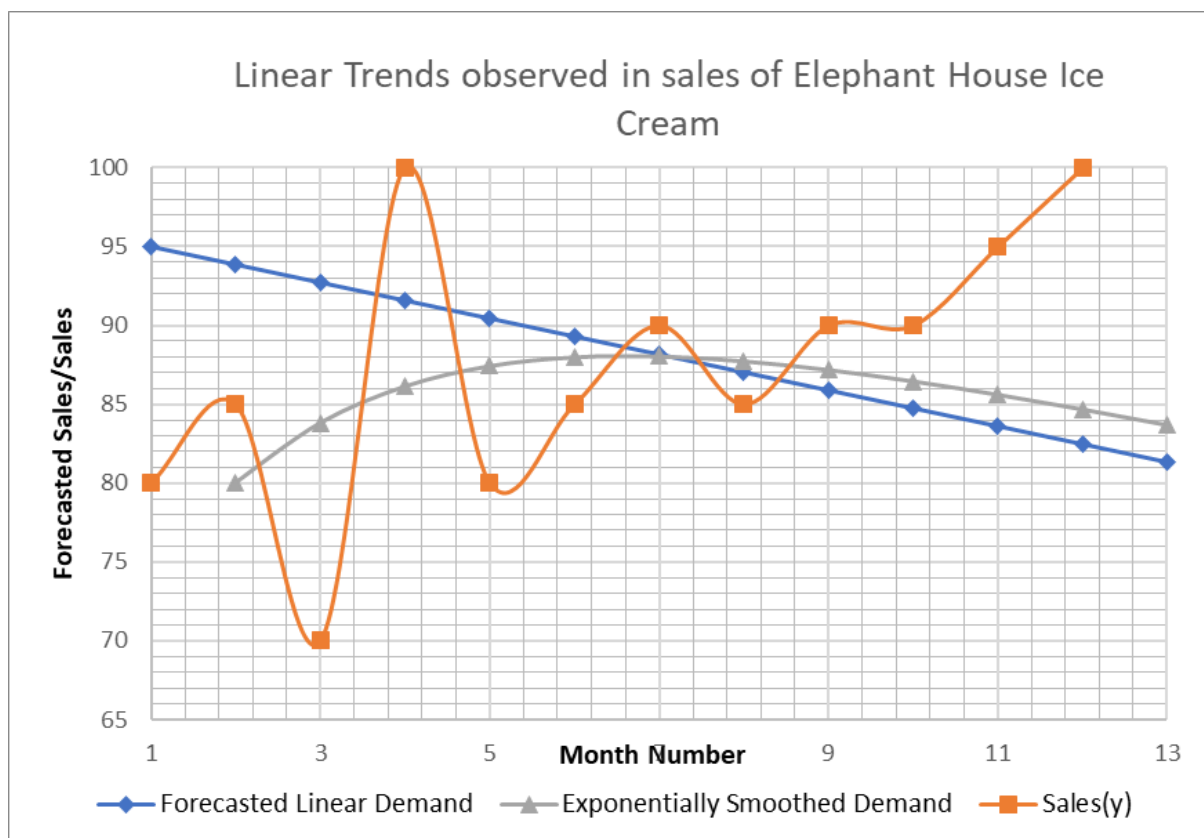
| Category                | Item           | September | October | November | December | January | February | March | April | May  | June | July | August |
|-------------------------|----------------|-----------|---------|----------|----------|---------|----------|-------|-------|------|------|------|--------|
| Ice cream               | Elephant house | 80        | 85      | 70       | 100      | 80      | 85       | 90    | 85    | 90   | 90   | 95   | 100    |
|                         | Cargills       | 40        | 38      | 40       | 60       | 60      | 50       | 50    | 40    | 40   | 35   | 35   | 30     |
| Perfume                 | Men's          | 10        | 10      | 10       | 10       | 10      | 10       | 10    | 10    | 10   | 10   | 10   | 10     |
|                         | Women's        | 10        | 10      | 10       | 10       | 10      | 10       | 10    | 10    | 10   | 10   | 10   | 10     |
| Chicken                 | Bairaha        | 20        | 20      | 20       | 35       | 20      | 20       | 20    | 35    | 20   | 20   | 20   | 20     |
| Sausage                 |                | 30        | 30      | 30       | 50       | 30      | 30       | 30    | 50    | 30   | 30   | 30   | 30     |
| Party items             |                | 5         | 5       | 5        | 5        | 5       | 5        | 5     | 5     | 5    | 5    | 5    | 5      |
| Stationary              | Atlas CR       | 40        | 40      | 40       | 40       | 40      | 40       | 40    | 40    | 40   | 40   | 40   | 40     |
| Daily medicine          | Siddalepa      | 15        | 15      | 15       | 15       | 15      | 15       | 15    | 15    | 15   | 15   | 15   | 15     |
| Toilet paper            | Flora          | 25        | 25      | 25       | 25       | 25      | 25       | 25    | 25    | 25   | 25   | 25   | 25     |
| Incense sticks          | Amritha        | 50        | 50      | 50       | 50       | 50      | 50       | 50    | 50    | 50   | 50   | 50   | 50     |
| Bulbs                   | Phillips       | 6         | 6       | 6        | 6        | 6       | 6        | 6     | 6     | 6    | 6    | 6    | 6      |
| Wicks                   |                | 50        | 50      | 50       | 50       | 50      | 50       | 50    | 50    | 50   | 50   | 50   | 50     |
| Washing powder          | Sunlight       | 100       | 100     | 100      | 100      | 100     | 100      | 100   | 100   | 100  | 100  | 100  | 100    |
| Washing items           | Dettol         | 15        | 15      | 15       | 15       | 15      | 15       | 15    | 15    | 15   | 15   | 15   | 15     |
| Liquid wash             | Vim            | 60        | 60      | 60       | 60       | 60      | 60       | 60    | 60    | 60   | 60   | 60   | 60     |
| Mosquito repellent Coil | Ninja          | 10        | 10      | 10       | 10       | 10      | 10       | 10    | 10    | 10   | 10   | 10   | 10     |
|                         | Mortein        | 30        | 30      | 30       | 30       | 30      | 30       | 30    | 30    | 30   | 30   | 30   | 30     |
| Toothpaste              | Signal         | 60        | 60      | 60       | 60       | 60      | 60       | 60    | 60    | 60   | 60   | 60   | 60     |
|                         | Clogard        | 15        | 15      | 15       | 15       | 15      | 15       | 15    | 15    | 15   | 15   | 15   | 15     |
| Salmon                  | Jack mackarel  | 10        | 10      | 10       | 10       | 10      | 10       | 10    | 10    | 10   | 10   | 10   | 10     |
| Soya                    | Raigam         | 25        | 25      | 25       | 25       | 25      | 25       | 25    | 25    | 25   | 25   | 25   | 25     |
|                         | Lankasoy       | 25        | 25      | 25       | 25       | 25      | 25       | 25    | 25    | 25   | 25   | 25   | 25     |
| Potatoes                |                | 30        | 30      | 30       | 30       | 30      | 30       | 30    | 30    | 30   | 30   | 30   | 30     |
| Garlic                  |                | 10        | 10      | 10       | 10       | 10      | 10       | 10    | 10    | 10   | 10   | 10   | 10     |
| Onions                  |                | 60        | 60      | 60       | 60       | 60      | 60       | 60    | 60    | 60   | 60   | 60   | 60     |
| Pasta                   | Harischandra   | 10        | 10      | 10       | 10       | 10      | 10       | 10    | 10    | 10   | 10   | 10   | 10     |
| Noodles                 | Kottu mee      | 160       | 160     | 160      | 160      | 160     | 160      | 160   | 160   | 160  | 160  | 160  | 160    |
|                         | Harischandra   | 20        | 20      | 20       | 20       | 20      | 20       | 20    | 20    | 20   | 20   | 20   | 20     |
| Lunch sheet             |                | 60        | 60      | 60       | 60       | 60      | 60       | 60    | 60    | 60   | 60   | 60   | 60     |
| Soap                    | Lux            | 20        | 20      | 20       | 20       | 20      | 20       | 20    | 20    | 20   | 20   | 20   | 20     |
| Handwash                | Lifebuoy       | 15        | 15      | 15       | 15       | 15      | 15       | 15    | 15    | 15   | 15   | 15   | 15     |
| Envelopes               |                | 30        | 30      | 30       | 30       | 30      | 30       | 30    | 30    | 30   | 30   | 30   | 30     |
| Hair-gel                |                | 5         | 5       | 5        | 5        | 5       | 5        | 5     | 5     | 5    | 5    | 5    | 5      |
| Hair dye                |                | 20        | 20      | 20       | 20       | 20      | 20       | 20    | 20    | 20   | 20   | 20   | 20     |
| Face powder             |                | 3         | 3       | 3        | 3        | 3       | 3        | 3     | 3     | 3    | 3    | 3    | 3      |
| Sanitary products       | Fems           | 75        | 75      | 75       | 75       | 75      | 75       | 75    | 75    | 75   | 75   | 75   | 75     |
| Sweets                  | Ritzbury       | 100       | 100     | 100      | 100      | 100     | 100      | 100   | 100   | 100  | 100  | 100  | 100    |
|                         | Uswatte        | 50        | 50      | 50       | 50       | 50      | 50       | 50    | 50    | 50   | 50   | 50   | 50     |
| Shampoo                 | Sunsilk        | 30        | 30      | 30       | 30       | 30      | 30       | 30    | 30    | 30   | 30   | 30   | 30     |
|                         | Dandex         | 10        | 10      | 10       | 10       | 10      | 10       | 10    | 10    | 10   | 10   | 10   | 10     |
| Powdered coconut milk   | Maggie         | 10        | 10      | 10       | 10       | 10      | 10       | 10    | 10    | 10   | 10   | 10   | 10     |
| Soup cubes              | Knorr          | 30        | 30      | 30       | 30       | 30      | 30       | 30    | 30    | 30   | 30   | 30   | 30     |
| Condiments              | Kist           | 15        | 15      | 15       | 15       | 15      | 15       | 15    | 15    | 15   | 15   | 15   | 15     |
| Culinary                | Mas            | 100       | 100     | 100      | 100      | 100     | 100      | 100   | 100   | 100  | 100  | 100  | 100    |
| Dried fish              |                | 15        | 15      | 15       | 15       | 15      | 15       | 15    | 15    | 15   | 15   | 15   | 15     |
| Eggs                    |                | 100       | 100     | 100      | 100      | 100     | 100      | 100   | 100   | 100  | 100  | 100  | 100    |
| Razor                   | Bic            | 90        | 90      | 90       | 90       | 90      | 90       | 90    | 90    | 90   | 90   | 90   | 90     |
| Biscuits                | Maliban        | 120       | 120     | 120      | 120      | 120     | 120      | 120   | 120   | 120  | 120  | 120  | 120    |
|                         | Munchee        | 70        | 70      | 70       | 70       | 70      | 70       | 70    | 70    | 70   | 70   | 70   | 70     |
| Slippers                |                | 10        | 10      | 10       | 10       | 10      | 10       | 10    | 10    | 10   | 10   | 10   | 10     |
| Bakery items            | Flour          | 50        | 50      | 50       | 50       | 50      | 50       | 50    | 50    | 50   | 50   | 50   | 50     |
|                         | Motha          | 60        | 60      | 60       | 60       | 60      | 60       | 60    | 60    | 60   | 60   | 60   | 60     |
| Soft drinks             | Elephant house | 250       | 250     | 250      | 500      | 250     | 250      | 250   | 350   | 250  | 250  | 250  | 250    |
| Tea                     | Zesta          | 150       | 150     | 150      | 150      | 150     | 150      | 150   | 150   | 150  | 150  | 150  | 150    |
| Milk powder             | Anchor         | 55        | 55      | 55       | 55       | 55      | 55       | 55    | 70    | 55   | 55   | 55   | 55     |
|                         | Pelwatta       | 48        | 48      | 48       | 48       | 48      | 48       | 48    | 60    | 48   | 48   | 48   | 48     |
| Milk                    | Ambewela       | 60        | 60      | 60       | 60       | 60      | 60       | 60    | 60    | 60   | 60   | 60   | 60     |
| Toffee                  |                | 5000      | 5000    | 5000     | 5000     | 5000    | 5000     | 5000  | 5000  | 5000 | 5000 | 5000 | 5000   |
| Coconut                 |                | 250       | 250     | 250      | 250      | 250     | 250      | 250   | 250   | 250  | 250  | 250  | 250    |
| Brooms                  |                | 20        | 20      | 20       | 20       | 20      | 20       | 20    | 20    | 20   | 20   | 20   | 20     |
| Baked biscuits          |                | 30        | 30      | 30       | 30       | 30      | 30       | 30    | 30    | 30   | 30   | 30   | 30     |

Table 2: Sales data collected (Past months)

By using the above data and sales data of previous years, demand trends for some items are forecasted as follows. For this, least square method was used, and the resultant curve obtained after exponential smoothing is also drawn. [3]

### Increasing Trend for Ice cream Elephant house

|           |                |          |         |      |                |                          | b | 0.323575 |
|-----------|----------------|----------|---------|------|----------------|--------------------------|---|----------|
|           |                |          |         |      |                |                          | a | 85.39676 |
| Month     | Purchase Order | Sales(y) | Time(x) | xy   | x <sup>2</sup> | Forecasted linear demand |   |          |
| September | 80             | 80       | 1       | 80   | 1              | 85.72033898              |   |          |
| October   | 85             | 85       | 2       | 170  | 4              | 86.04391371              |   |          |
| November  | 70             | 70       | 3       | 210  | 9              | 86.36748844              |   |          |
| December  | 100            | 100      | 4       | 400  | 16             | 86.69106317              |   |          |
| January   | 80             | 80       | 5       | 400  | 25             | 87.0146379               |   |          |
| February  | 85             | 85       | 6       | 510  | 36             | 87.33821263              |   |          |
| March     | 90             | 90       | 7       | 630  | 49             | 87.66178737              |   |          |
| April     | 85             | 85       | 8       | 680  | 64             | 87.9853621               |   |          |
| May       | 90             | 90       | 9       | 810  | 81             | 88.30893683              |   |          |
| June      | 90             | 90       | 10      | 900  | 100            | 88.63251156              |   |          |
| July      | 95             | 95       | 11      | 1045 | 121            | 88.95608629              |   |          |
| August    | 100            | 100      | 12      | 1200 | 144            | 89.27966102              |   |          |
|           |                | 1050     | 78      | 7035 | 650            | 110.6355932              |   |          |



*Figure 2: Trend for Elephant House ice cream*

Similarly, calculations were carried out with regards to items that showed increasing or decreasing trends.

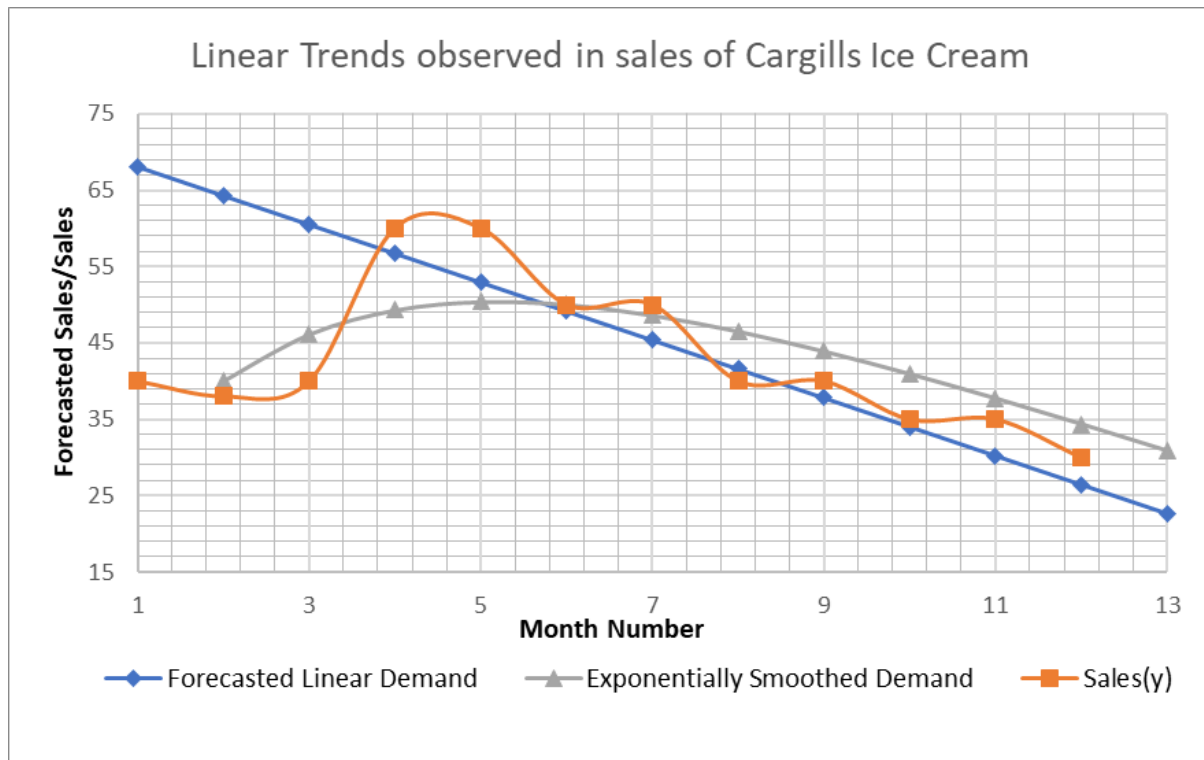


Figure 3: Trend for Cargills ice cream

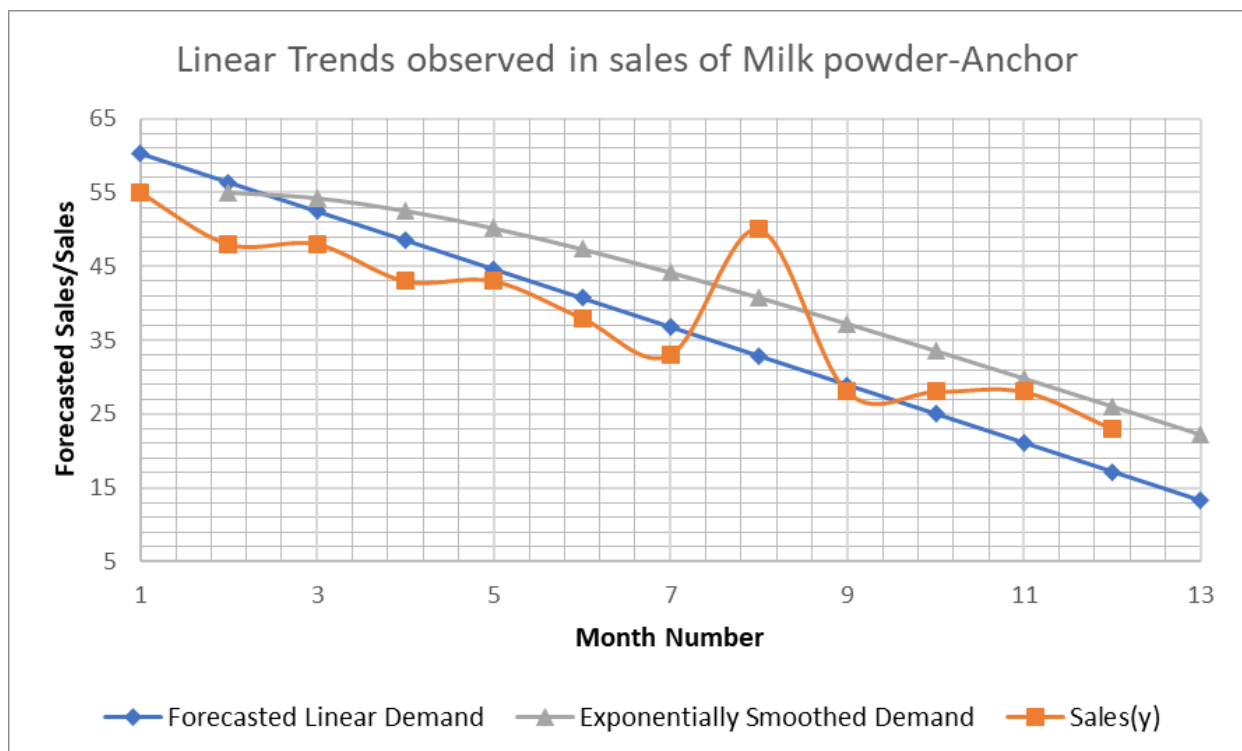


Figure 4: Trend for Anchor milk powder

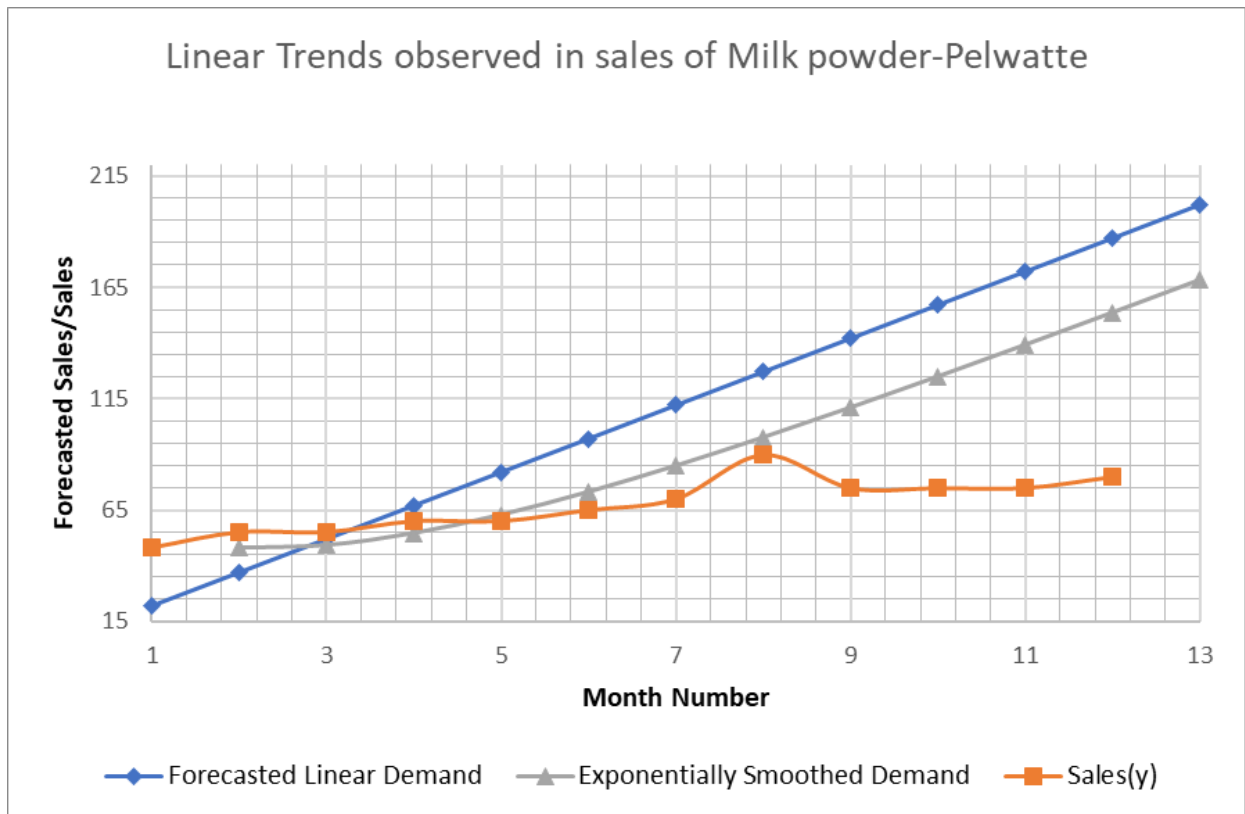


Figure 5: Trend for Pelwatte milk powder

From the data of the past three years (2018/19 – 2022/23) that was predicted after a few conversations with the owner, we forecasted the demand for the above items for the upcoming year (2023/24) including the seasonal factors to determine the economic order quantities for that year.

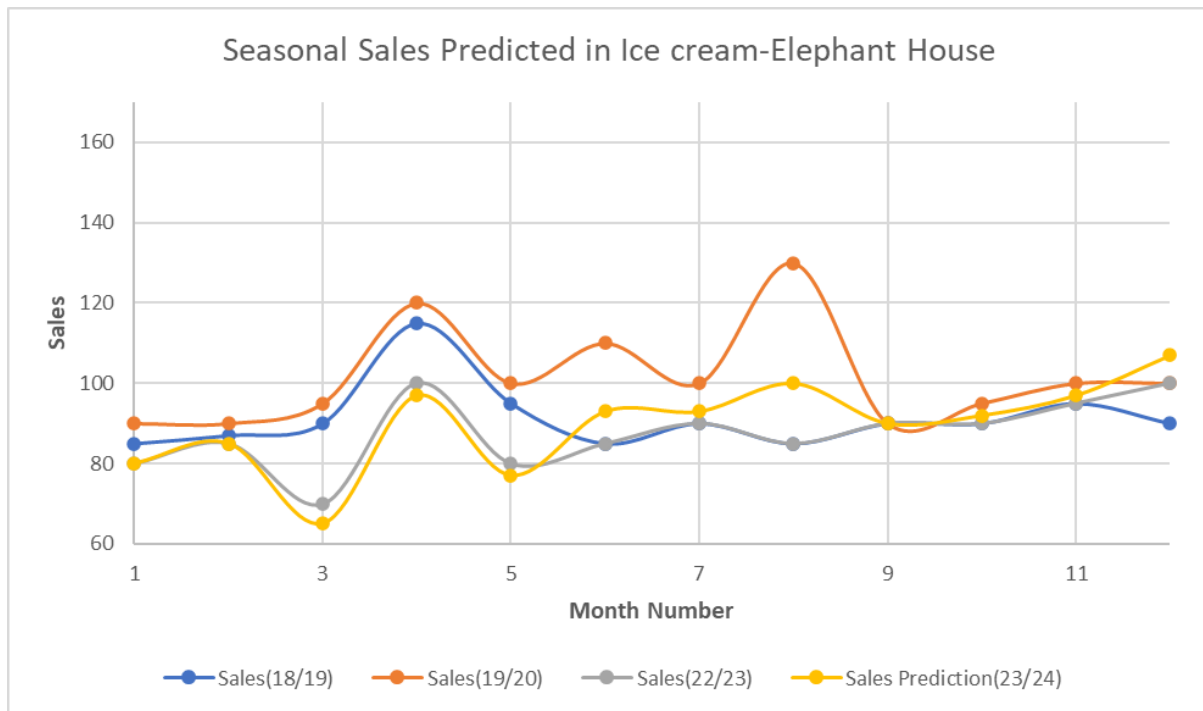


Figure 6: Sales prediction for Elephant House ice cream 2023/24

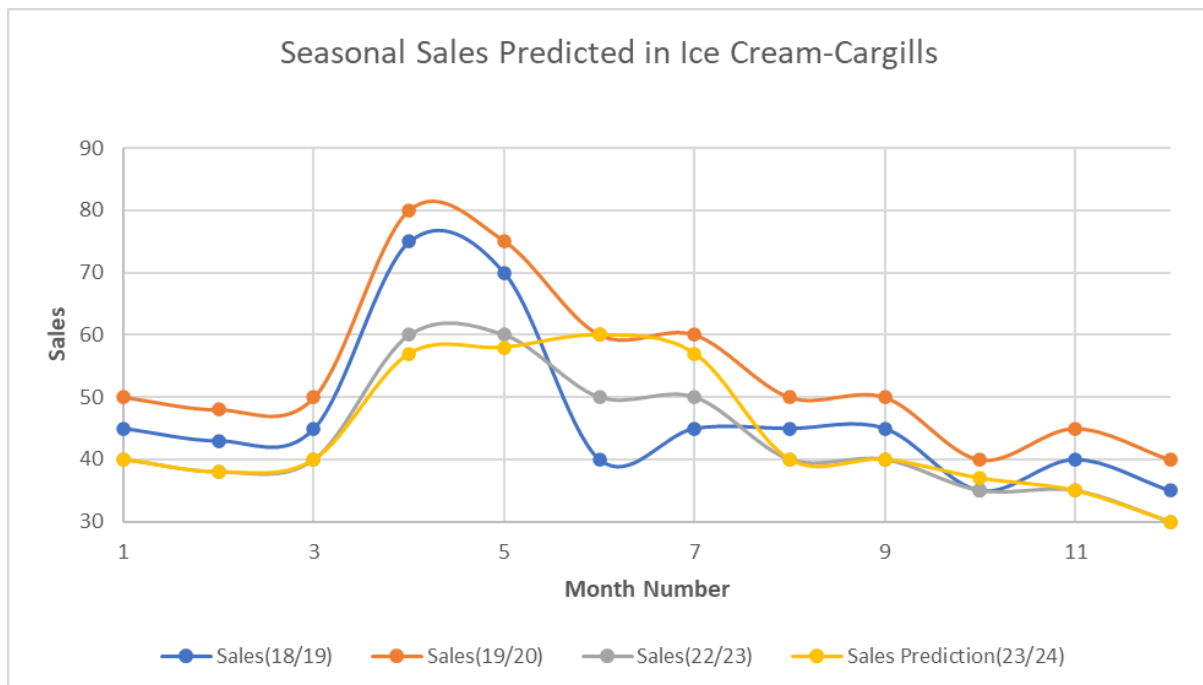


Figure 7: Sales prediction for Cargills ice cream 2023/24

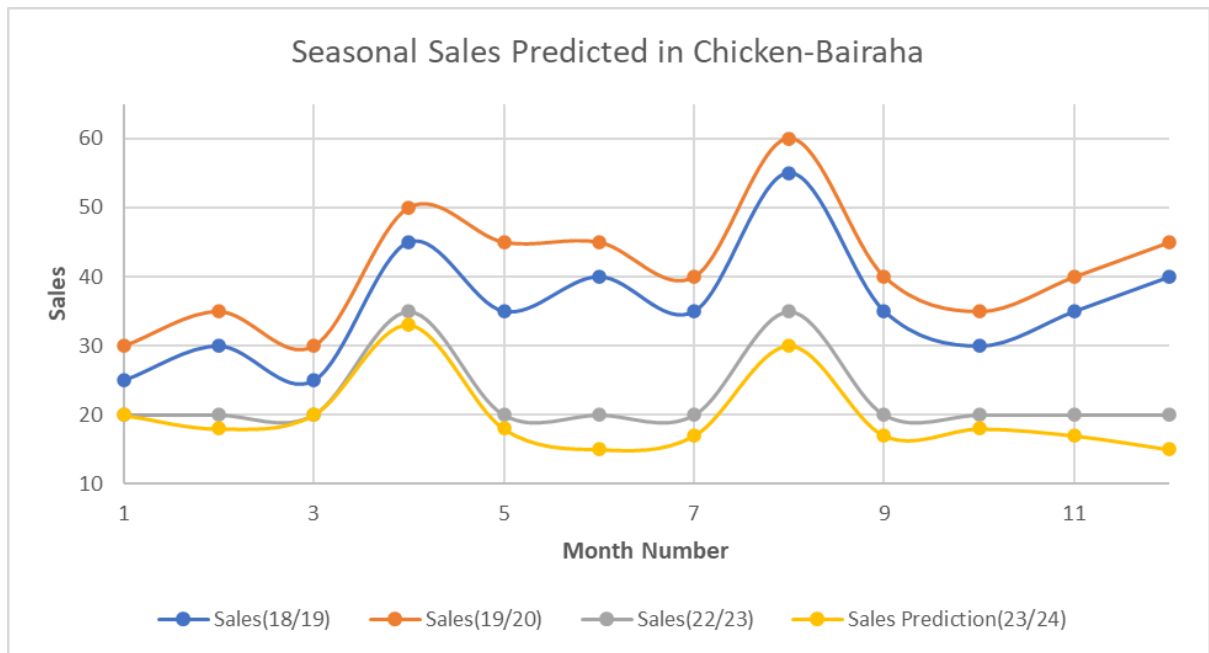


Figure 8: Sales prediction for Bairaha chicken

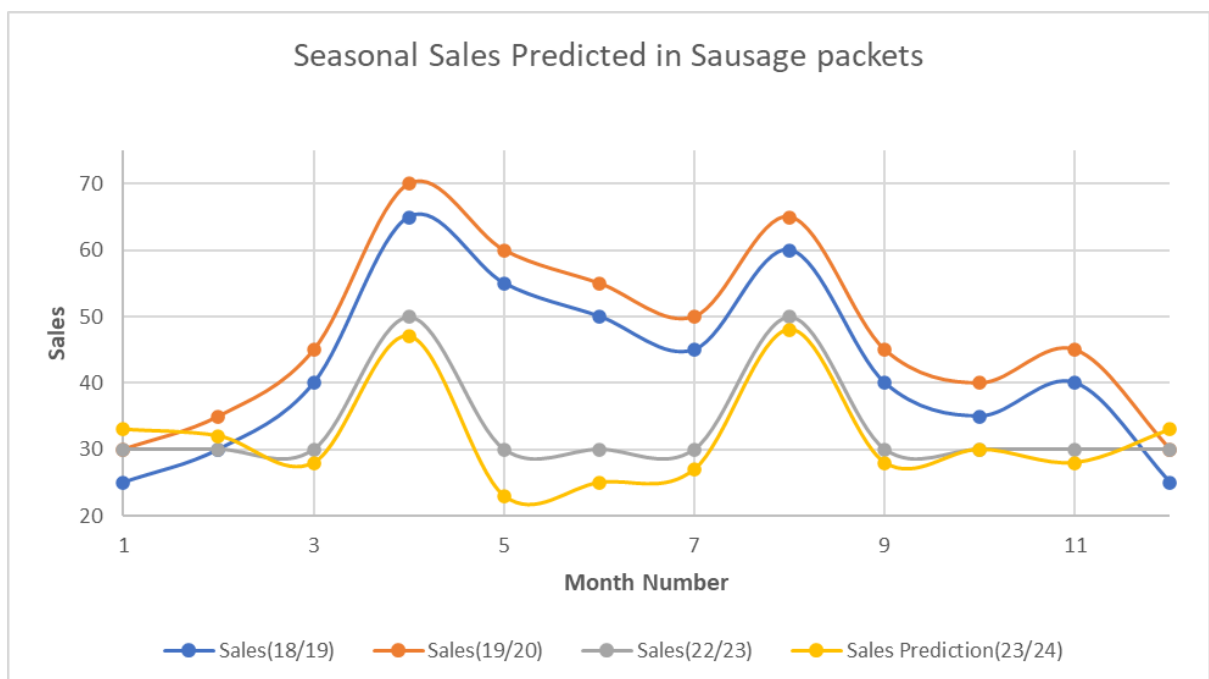


Figure 9: Sales prediction for Sausage packets 2023/24

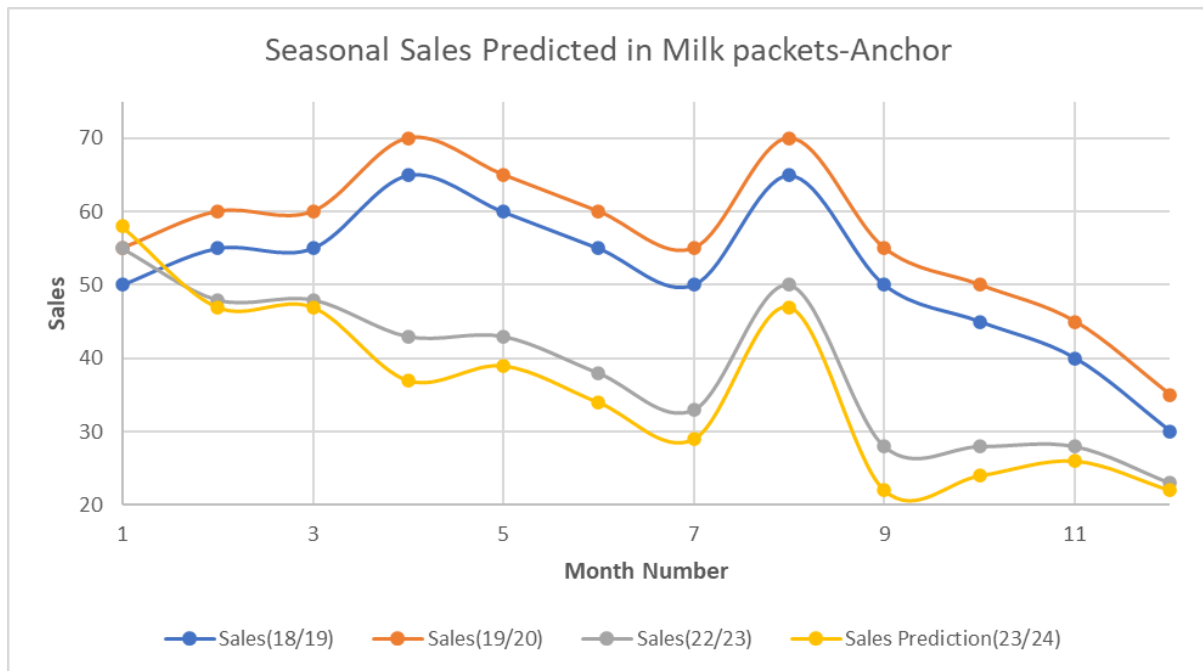


Figure 10: Sales predictions for Anchor milk packets

Based on the data we collected, the bill of material for this store is as follows.

| Category                | Item           | Category              | Item           |
|-------------------------|----------------|-----------------------|----------------|
| Ice cream               | Elephant house | Handwash              | Lifebuoy       |
|                         | Cargills       | Envelopes             |                |
| Perfume                 | Men's          | Hair-gel              |                |
|                         | Women's        | Hair dye              |                |
| Chicken                 | Bairaha        | Face powder           |                |
| Sausage                 |                | Sanitary products     | Fems           |
| Party items             |                | Sweets                | Ritzbury       |
| Stationary              | Atlas CR       |                       | Uswatte        |
| Daily medicine          | Siddalepa      | Shampoo               | Sunsilk        |
| Toilet paper            | Flora          |                       | Dandex         |
| Incense sticks          | Amritha        | Powdered coconut milk | Maggie         |
| Bulbs                   | Phillips       | Soup cubes            | Knorr          |
| Wicks                   |                | Condiments            | Kist           |
| Washing powder          | Sunlight       | Culinary              | Mas            |
| Washing items           | Dettol         | Dried fish            |                |
| Liquid wash             | Vim            | Eggs                  |                |
| Mosquito repellent Coil | Ninja          | Razor                 | Bic            |
|                         | Mortein        | Biscuits              | Maliban        |
| Toothpaste              | Signal         |                       | Munchee        |
|                         | Clogard        | Slippers              |                |
| Salmon                  | Jack mackarel  | Bakery items          | Flour          |
| Soya                    | Raigam         |                       | Motha          |
|                         | Lankasoy       | Soft drinks           | Elephant house |
| Potatoes                |                | Tea                   | Zesta          |
| Garlic                  |                | Milk powder           | Anchor         |
| Onions                  |                |                       | Pelwatte       |
| Pasta                   | Harischandra   | Milk                  | Ambewela       |
| Noodles                 | Kottu mee      | Toffee                |                |
|                         | Harischandra   | Coconut               |                |
| Lunch sheet             |                | Brooms                |                |
| Soap                    | Lux            | Baked biscuits        |                |

Table 3: Bill of materials

## 2.2 Facility Layout Planning

The existing layout is not an organized one. Most sold items are stores upfront while all other items are scattered in an unorganized manner. By using icograms online tool, a 3D model of the existing layout can be illustrated as follows. [4]



*Figure 11: Existing layout*

To improve the customer experience and reduce their waiting times, the current layout should be readjusted. It is impossible to plan the layout in a way that every customer buys their needs within a one step into the store. Hence, by analysing the data collected, we sorted out four most common lists of items bought by the customers who comes to the convenience store. These item lists are as follows.

| List 1(Lower Middle-Class Customers with Daily Wages) | List 2(Regular Upper Middle-Class Customers) | List 3(Customers Who Buy Goods Monthly/Weekly) | List 4(Daily Customers Who Comes at Peak Hours) |
|-------------------------------------------------------|----------------------------------------------|------------------------------------------------|-------------------------------------------------|
| Rice                                                  | Meat Products/Dried Fish/Salmon              | Rice                                           | Milk packets                                    |
| Potatoes/Onions                                       | Fresh Milk                                   | Flour                                          | Sweets/Bakery Items                             |
| Bread                                                 | Biscuits/Bakery Products                     | Dhal                                           | Stationary                                      |
| Biscuit                                               | Noodles/Pasta                                | Meat Products                                  | Ice cream                                       |
|                                                       | Ice Cream                                    | Biscuits                                       |                                                 |
|                                                       |                                              | Soap                                           |                                                 |
|                                                       |                                              | Milk Powder                                    |                                                 |
|                                                       |                                              | Tea                                            |                                                 |
|                                                       |                                              | Washing Powder                                 |                                                 |
|                                                       |                                              | Vim                                            |                                                 |
|                                                       |                                              | Toothpaste                                     |                                                 |
|                                                       |                                              | Noodles/Pasta                                  |                                                 |
|                                                       |                                              | Lunch sheets                                   |                                                 |
|                                                       |                                              | Ice Cream/Milk Products                        |                                                 |

*Figure 12: Four most common item lists used by customers.*

Based on these four item lists, we created three sample layouts which consist of different arrangements which the items in the above lists are accessible for customers within a minimum amount of time. In those layouts, following specialities are present.

- Layout 1: All refrigerators are placed close to one another for ease of maintenance and electric efficiency.



*Figure 13: Layout 1*

- Layout 2: Two counters are placed to reduce the customer rush.



*Figure 14: Layout 2*

- Layout 3: A separate counter is placed to weigh dry food items.



*Figure 15: Layout 3*

For each of these layouts, we generated the following table based on the duration it takes to retrieve each product in minutes. These durations were measured with consideration for customers walking from the entrance to the product. [5]

| Items               | Time taken for grabbing each product in minutes |                            |                            |                            |
|---------------------|-------------------------------------------------|----------------------------|----------------------------|----------------------------|
|                     | Time allocated in Existing Layout               | Time allocated in Layout 1 | Time allocated in Layout 2 | Time allocated in Layout 3 |
| Rice                | 2                                               | 2                          | 2                          | 3                          |
| Potatoes/Onions     | 2                                               | 2                          | 2                          | 3                          |
| Bread               | 0.5                                             | 0.5                        | 0.5                        | 0.5                        |
| Salmon              | 1                                               | 1                          | 1                          | 1                          |
| Biscuit             | 0.2                                             | 0.2                        | 0.2                        | 0.2                        |
| Dried Fish          | 2.5                                             | 2.5                        | 2.5                        | 2.5                        |
| Meat Products       | 0.5                                             | 2                          | 2                          | 2                          |
| Fresh Milk          | 0.5                                             | 0.5                        | 0.5                        | 0.5                        |
| Noodles/Pasta       | 1.5                                             | 1.5                        | 1.5                        | 1.5                        |
| Ice Cream           | 0.5                                             | 0.8                        | 0.8                        | 0.8                        |
| Rice                | 2                                               | 2                          | 2                          | 3                          |
| Flour               | 1                                               | 1                          | 1                          | 1                          |
| Dhal                | 2                                               | 2                          | 2                          | 3                          |
| Biscuits            | 0.5                                             | 0.5                        | 0.5                        | 0.5                        |
| Soap                | 1                                               | 1                          | 1                          | 1                          |
| Milk Powder         | 1                                               | 1                          | 1                          | 1                          |
| Tea                 | 1.5                                             | 1.5                        | 1.5                        | 1.5                        |
| Washing Powder      | 1.5                                             | 1.5                        | 1.5                        | 1.5                        |
| Vim                 | 1.5                                             | 1.5                        | 1.5                        | 1.5                        |
| Toothpaste          | 1.5                                             | 1.5                        | 1.5                        | 1.5                        |
| Lunch sheets        | 1.5                                             | 1.5                        | 1.5                        | 1.5                        |
| Sweets/Bakery Items | 0.5                                             | 0.5                        | 0.5                        | 0.5                        |
| Stationary          | 1                                               | 1                          | 1                          | 1                          |
| Cashier Time        | 0.5                                             | 0.5                        | 0.5                        | 0.4                        |

*Table 4: Time taken for grabbing products based on layouts.*

## 2.2.1 Simulation Procedure

### 2.2.1.1 Simulation using Excel.

This was mainly carried out by identifying different item lists shopped by different categories of customers, item classifications and proposed layouts.

We initially identified the four primary good categories, which are:

- Frozen foods.
- Dry foods near the entrance.
- Dry foods away from the entrance.
- Goods that need measuring.

We measured the amount of time it took to pick up each product in minutes, both from the entrance and while moving from one product to another.

|                                                                      | Time(min) |
|----------------------------------------------------------------------|-----------|
| Entrance to the frozen foods                                         | 0.5       |
| Entrance to the dry foods near entrance                              | 1         |
| Entrance to the dry foods away from entrance                         | 1.5       |
| Foods that need measuring                                            | 2         |
| From frozen foods to dry foods near the entrance                     | 0.5       |
| From dry foods near the entrance to the dry foods away from entrance | 0.5       |
| From dry foods away from the entrance to foods that need measuring   | 0.5       |
| From frozen foods to the dry foods away from the entrance            | 1         |
| From frozen foods to the foods that need measuring                   | 1.5       |
| From dry foods near the entrance to the foods that need measuring    | 1         |

*Table 5: Time taken for changing between items.*

Let's assume,

|                                                                  |      |
|------------------------------------------------------------------|------|
| Cashier time for one item                                        | 0.5  |
| Cashier time with two cashiers                                   | 0.25 |
| Cashier time with separate station for foods that need measuring | 0.4  |

*Table 6: Cashier waiting times.*

It should be noted that Cashier time with separate station for foods that need measuring is decreased due to measuring activities carried out in a separate space and cashier que time not taken up for that task.

Afterwards, we computed the time it takes for a single customer to pick up different products in minutes, both in the current layout and in alternative layouts. The resulting values are as follows,

|   | Time(min) in Existing Layout | Time(min) in Layout 1 | Time(min) in Layout 2 | Time(min) in Layout 3 |
|---|------------------------------|-----------------------|-----------------------|-----------------------|
| A | 4                            | 3.75                  | 3                     | 3.9                   |
| B | 3                            | 2.75                  | 2.25                  | 3                     |
| C | 3.5                          | 3.25                  | 2.75                  | 3.4                   |
| D | 2                            | 1.75                  | 1.5                   | 2                     |
| E | 3.5                          | 3.25                  | 2.75                  | 3.4                   |
| F | 2.5                          | 2.25                  | 2                     | 2.5                   |
| G | 3                            | 2.75                  | 2.5                   | 2.9                   |
| H | 1                            | 0.75                  | 0.75                  | 1                     |
| I | 3.5                          | 3.5                   | 2.75                  | 3.4                   |
| J | 2.5                          | 2.5                   | 2                     | 2.5                   |
| K | 3                            | 3                     | 2.5                   | 2.9                   |
| L | 1.5                          | 1.5                   | 1.25                  | 1.5                   |
| M | 3                            | 3                     | 2.5                   | 2.9                   |
| N | 2                            | 2                     | 1.75                  | 2                     |
| O | 2.5                          | 2.5                   | 2.25                  | 2.9                   |

*Table 7: Times for grabbing different products in different layouts.*

|   |                                                                                        |
|---|----------------------------------------------------------------------------------------|
| A | Frozen + Dry foods near entrance + Dry foods away entrance + Foods that need measuring |
| B | Frozen + Dry foods near entrance + Dry foods away entrance                             |
| C | Frozen + Dry foods near entrance + Foods that need measuring                           |
| D | Frozen + Dry foods near entrance                                                       |
| E | Frozen + Dry foods away entrance + Foods that need measuring                           |
| F | Frozen + Dry foods away entrance                                                       |
| G | Frozen + Foods that need measuring                                                     |
| H | Frozen                                                                                 |
| I | Dry foods near entrance + Dry foods away entrance + Foods that need measuring          |
| J | Dry foods near entrance + Dry foods away entrance                                      |
| K | Dry foods near entrance + Foods that need measuring                                    |
| L | Dry foods near entrance                                                                |
| M | Foods that need measuring                                                              |
| N | Dry foods away entrance                                                                |
| O | Foods that need measuring                                                              |

*Table 8: Combinations of items considered in table 5.*

We can interpret these values in a graph.

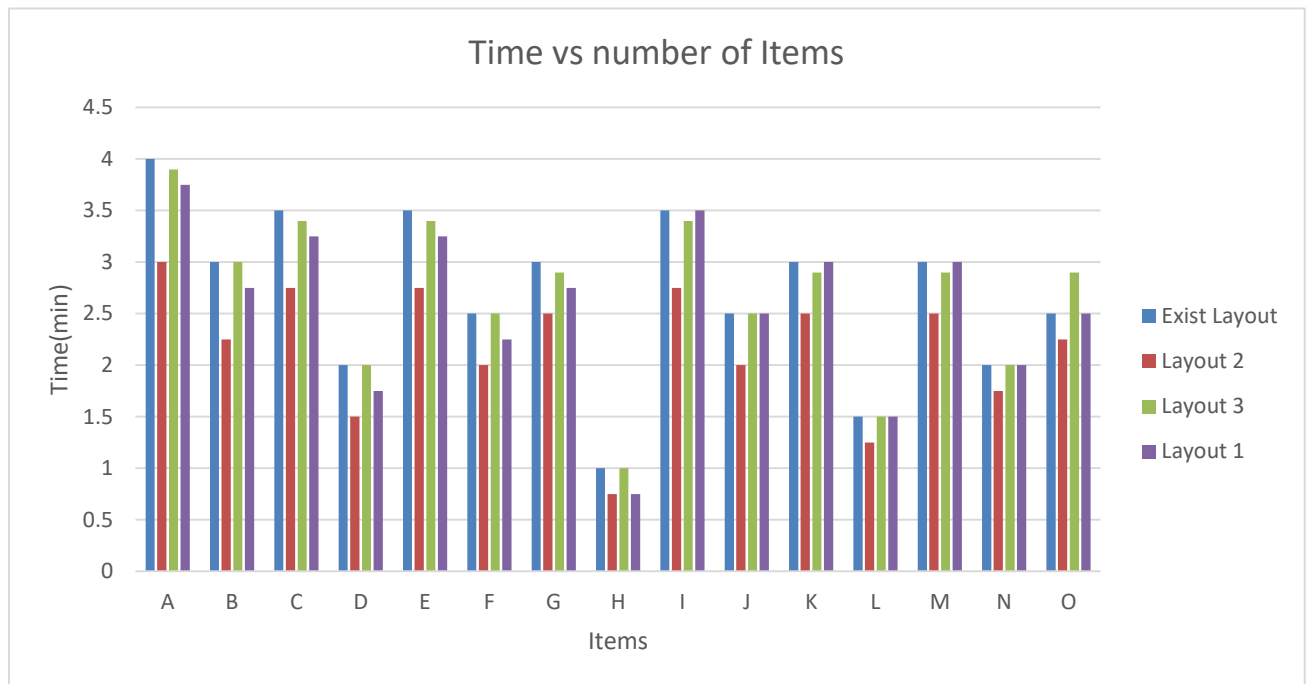


Figure 16: Graphs of Time Vs Layout.

### 2.2.1.2 Simulation using Tecnomatix software [6]

We utilized Tecnomatix software to create a simulated model represented by a tree diagram.

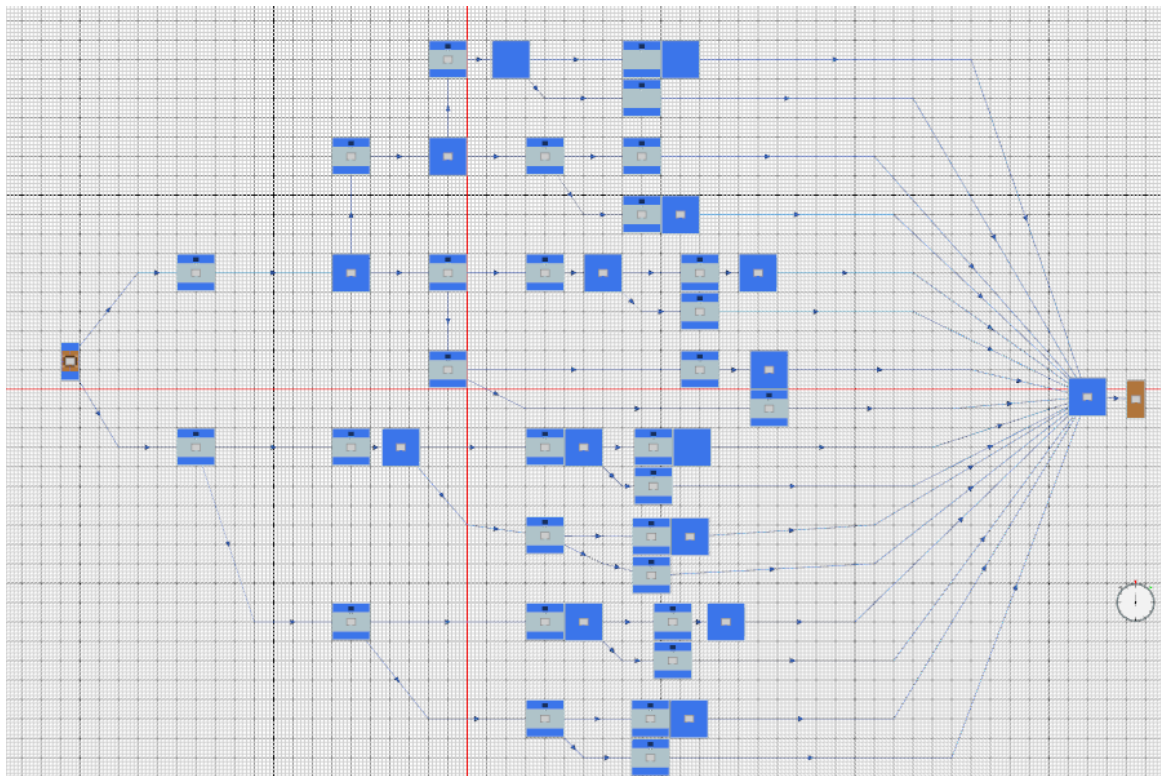


Figure 17: Simulation of the layout.

Stations are employed to guide customers in selecting products based on their preferences. Buffers are employed to introduce a delay that corresponds to the time it takes to select products via different routes.

We adjusted the delay times in buffers and at the cashiers based on both our current layout and alternative layouts.

Next, we determined the number of entries into the drain for a specified period (12 hours) using various layouts.

## **Results Obtained**

### ➤ Existing Layout

#### **Material Flow Properties**

| Object | Number of Entries | Number of Exits | Minimum Contents | Maximum Contents | Relative Empty | Relative Full | Relative Occupation without Interruptions | Relative Occupation with Interruptions |
|--------|-------------------|-----------------|------------------|------------------|----------------|---------------|-------------------------------------------|----------------------------------------|
| Drain  | 1230              | 1230            | 0                | 1                | 100.00%        | -             | 0.00%                                     | 0.00%                                  |

### ➤ Layout 1

#### **Material Flow Properties**

| Object | Number of Entries | Number of Exits | Minimum Contents | Maximum Contents | Relative Empty | Relative Full | Relative Occupation without Interruptions | Relative Occupation with Interruptions |
|--------|-------------------|-----------------|------------------|------------------|----------------|---------------|-------------------------------------------|----------------------------------------|
| Drain  | 1550              | 1550            | 0                | 1                | 100.00%        | -             | 0.00%                                     | 0.00%                                  |

### ➤ Layout 2

#### **Material Flow Properties**

| Object | Number of Entries | Number of Exits | Minimum Contents | Maximum Contents | Relative Empty | Relative Full | Relative Occupation without Interruptions | Relative Occupation with Interruptions |
|--------|-------------------|-----------------|------------------|------------------|----------------|---------------|-------------------------------------------|----------------------------------------|
| Drain  | 1880              | 1880            | 0                | 1                | 100.00%        | -             | 0.00%                                     | 0.00%                                  |

### ➤ Layout 3

#### **Material Flow Properties**

| Object | Number of Entries | Number of Exits | Minimum Contents | Maximum Contents | Relative Empty | Relative Full | Relative Occupation without Interruptions | Relative Occupation with Interruptions |
|--------|-------------------|-----------------|------------------|------------------|----------------|---------------|-------------------------------------------|----------------------------------------|
| Drain  | 1670              | 1670            | 0                | 1                | 100.00%        | -             | 0.00%                                     | 0.00%                                  |

Considering the simulation results and other factors, we opted for **layout 2**.

## 2.3 Inventory Control [7]

The current state of the inventory is as follows.

- No formal way of inventory or bookkeeping is maintained.
- Goods are stocked on a relevant basis.
- Number of stocks needed is estimated based on experience.
- Number of sales made per day is a rough estimate.

### 2.3.1 ABC Analysis

- An ABC Analysis was performed to find out the values of products sold.
- It was identified that all items sold are of lesser value than that of the total income (Category C).

| Category                | Item           | Maximum sales done in the year | Unit Price(LKR) | Total Price(LKR) | Category   |
|-------------------------|----------------|--------------------------------|-----------------|------------------|------------|
| Ice cream               | Elephant house | 100                            | 200             | 20000            | Category C |
|                         | Cargills       | 60                             | 180             | 10800            | Category C |
| Perfume                 | Men's          | 10                             | 500             | 5000             | Category C |
|                         | Women's        | 10                             | 500             | 5000             | Category C |
| Chicken                 | Bairaha(kg)    | 35                             | 1870            | 65450            | Category C |
| Sausage                 | (packs)        | 50                             | 612             | 30600            | Category C |
| Party items             |                | 5                              | 400             | 2000             | Category C |
| Stationary              | Atlas CR       | 40                             | 430             | 17200            | Category C |
| Daily medicine          | Siddalepa      | 15                             | 7.25            | 108.75           | Category C |
| Toilet paper            | Flora          | 25                             | 192             | 4800             | Category C |
| Incense sticks          | Amritha        | 50                             | 150             | 7500             | Category C |
| bulbs                   | Phillips       | 6                              | 925             | 5550             | Category C |
| Wicks                   |                | 50                             | 24              | 1200             | Category C |
| washing powder          | sunlight       | 100                            | 430             | 43000            | Category C |
| Hand Wash               | detol          | 15                             | 415             | 6225             | Category C |
| Liquid wash             | Vim            | 60                             | 400             | 24000            | Category C |
| Mosquito repellent Coil | Ninja          | 10                             | 170             | 1700             | Category C |
|                         | Mortein        | 30                             | 350             | 10500            | Category C |
| Toothpaste              | signal         | 60                             | 250             | 15000            | Category C |
|                         | clogard        | 15                             | 125             | 1875             | Category C |
| salmon                  | jack mackarel  | 10                             | 850             | 8500             | Category C |
| soya                    | raigam         | 25                             | 170             | 4250             | Category C |
|                         | lankasoy       | 25                             | 139             | 3475             | Category C |
| Potatoes                | (kg)           | 30                             | 330             | 9900             | Category C |
| Garlic                  | (kg)           | 10                             | 890             | 8900             | Category C |
| Onions                  | (kg)           | 60                             | 310             | 18600            | Category C |
| Pasta                   | Harischandra   | 10                             | 715             | 7150             | Category C |
| Noodles                 | Kottu mee      | 160                            | 130             | 20800            | Category C |
|                         | Harischandra   | 20                             | 340             | 6800             | Category C |
| Lunch sheet             | (packs)        | 60                             | 445             | 26700            | Category C |
| soap                    | lux            | 20                             | 175             | 3500             | Category C |

|                               |                |      |      |         |            |
|-------------------------------|----------------|------|------|---------|------------|
| handwash                      | lifebuoy       | 15   | 525  | 7875    | Category C |
| envelopes                     |                | 30   | 30   | 900     | Category C |
| Hair-gel                      |                | 5    | 420  | 2100    | Category C |
| hair dye                      |                | 20   | 280  | 5600    | Category C |
| face powder                   |                | 3    | 480  | 1440    | Category C |
| sanitary products             | Fems           | 75   | 150  | 11250   | Category C |
| sweets                        | Ritzbury       | 100  | 200  | 20000   | Category C |
|                               | Uswatte        | 50   | 80   | 4000    | Category C |
| shampoo                       | Sunsilk        | 30   | 785  | 23550   | Category C |
|                               | Dandex         | 10   | 540  | 5400    | Category C |
| podererd                      |                |      |      |         |            |
| coconut milk                  | maggie         | 10   | 2300 | 23000   | Category C |
| soup cubes                    | knorr          | 30   | 229  | 6870    | Category C |
| condiments                    | kist           | 15   | 600  | 9000    | Category C |
| culinary                      | mas(kg)        | 100  | 480  | 48000   | Category C |
| dried fish                    | (kg)           | 15   | 5280 | 79200   | Category C |
| eggs                          |                | 100  | 550  | 55000   | Category C |
| razor                         | Bic            | 90   | 110  | 9900    | Category C |
| biscuits                      | maliban        | 120  | 200  | 24000   | Category C |
|                               | munchee        | 70   | 200  | 14000   | Category C |
| slippers                      |                | 10   | 750  | 7500    | Category C |
| bakery items                  | flour(kg)      | 50   | 410  | 20500   | Category C |
|                               | motha          | 60   | 440  | 26400   | Category C |
|                               |                |      |      |         |            |
| soft drinks                   | elephant house | 500  | 250  | 125000  | Category C |
| tea                           | zesta          | 150  | 351  | 52650   | Category C |
| milk powder                   | anchor         | 70   | 1080 | 75600   | Category C |
|                               | Pelwatte       | 60   | 999  | 59940   | Category C |
| milk                          | ambewela       | 60   | 500  | 30000   | Category C |
| toffee                        | (Rs)           | 5000 | 1    | 5000    | Category C |
| coconut                       |                | 250  | 105  | 26250   | Category C |
| brooms                        |                | 20   | 1500 | 30000   | Category C |
| baked biscuits                |                | 30   | 300  | 9000    | Category C |
| Total Price of all items sold |                |      |      | 1215009 |            |

*Table 9: ABC Analysis Results.*

### 2.3.2 EOQ Calculation

An EOQ calculation for each product was done due to the following suitability understood for a small-scale retail store.

- EOQ model enhances cost efficiency for small retailers.
- It aids in managing cash flow effectively, crucial for businesses with limited cash resources.
- The model minimizes storage costs by ensuring efficient inventory turnover.
- It provides a basis for negotiations with suppliers, potentially leading to better pricing and terms.
- Small retailers benefit from reduced waste and associated costs by avoiding overstocking perishable items.

| Category                | Item           | Annual Demand | Annual Holding Cost | Annual Ordering Cost | EOQ  |
|-------------------------|----------------|---------------|---------------------|----------------------|------|
| Ice cream               | Elephant house | 1050          | 2000                | 1000                 | 32   |
|                         | Cargills       | 518           | 2000                | 1000                 | 23   |
| Perfume                 | Men's          | 120           | NA                  | 1000                 | NA   |
|                         | Women's        | 120           | NA                  | 1000                 | NA   |
| Chicken                 | Bairaha        | 270           | 2000                | 1000                 | 16   |
| Sausage                 |                | 400           | 2000                | 1000                 | 20   |
| Party items             |                | 60            | NA                  | 1000                 | NA   |
| Stationary              | Atlas CR       | 480           | 100                 | 1000                 | 98   |
| Daily medicine          | Siddalepa      | 180           | 100                 | 1000                 | 60   |
| Toilet paper            | Flora          | 300           | 100                 | 1000                 | 77   |
| Incense sticks          | Amritha        | 600           | 100                 | 1000                 | 110  |
| Bulbs                   | Phillips       | 72            | 100                 | 1000                 | 38   |
| Wicks                   |                | 600           | 100                 | 1000                 | 110  |
| Washing powder          | Sunlight       | 1200          | 100                 | 1000                 | 155  |
| Washing items           | Dettol         | 180           | 100                 | 1000                 | 60   |
| Liquid wash             | Vim            | 720           | 100                 | 1000                 | 120  |
| Mosquito repellent Coil | Ninja          | 120           | 100                 | 1000                 | 49   |
|                         | Mortein        | 360           | 100                 | 1000                 | 85   |
| Toothpaste              | Signal         | 720           | 100                 | 1000                 | 120  |
|                         | Clogard        | 180           | 100                 | 1000                 | 60   |
| Salmon                  | Jack mackarel  | 120           | 100                 | 1000                 | 49   |
| Soya                    | Raigam         | 300           | 100                 | 1000                 | 77   |
|                         | Lankasoy       | 300           | 100                 | 1000                 | 77   |
| Potatoes                |                | 360           | 500                 | 1000                 | 38   |
| Garlic                  |                | 120           | 500                 | 1000                 | 22   |
| Onions                  |                | 720           | 500                 | 1000                 | 54   |
| Pasta                   | Harischandra   | 120           | 100                 | 1000                 | 49   |
| Noodles                 | Kottu mee      | 1920          | 100                 | 1000                 | 196  |
|                         | Harischandra   | 240           | 100                 | 1000                 | 69   |
| Lunch sheet             |                | 720           | 100                 | 1000                 | 120  |
| Soap                    | Lux            | 240           | 500                 | 1000                 | 31   |
| Handwash                | Lifebuoy       | 180           | 100                 | 1000                 | 60   |
| Envelopes               |                | 360           | 100                 | 1000                 | 85   |
| Hair-gel                |                | 60            | 100                 | 1000                 | 35   |
| Hair dye                |                | 240           | 100                 | 1000                 | 69   |
| Face powder             |                | 36            | 100                 | 1000                 | 27   |
| Sanitary products       | Fems           | 900           | 100                 | 1000                 | 134  |
| Sweets                  | Ritzbury       | 1200          | 1000                | 1000                 | 49   |
|                         | Uswatte        | 600           | 1000                | 1000                 | 35   |
| Shampoo                 | Sunsilk        | 360           | 100                 | 1000                 | 85   |
|                         | Dandex         | 120           | 100                 | 1000                 | 49   |
| Powdered coconut milk   | Maggie         | 120           | 100                 | 1000                 | 49   |
| Soup cubes              | Knorr          | 360           | 100                 | 1000                 | 85   |
| Condiments              | Kist           | 180           | 100                 | 1000                 | 60   |
| Culinary                | Mas            | 1200          | 100                 | 1000                 | 155  |
| Dried fish              |                | 180           | 100                 | 1000                 | 60   |
| Eggs                    |                | 1200          | 100                 | 1000                 | 155  |
| Razor                   | Bic            | 1080          | 100                 | 1000                 | 147  |
| Biscuits                | Maliban        | 1440          | 100                 | 1000                 | 170  |
|                         | Munchee        | 840           | 100                 | 1000                 | 130  |
| Slippers                |                | 120           | 100                 | 1000                 | 49   |
| Bakery items            | Flour          | 600           | 100                 | 1000                 | 110  |
|                         | Motha          | 720           | 100                 | 1000                 | 120  |
| Soft drinks             | Elephant house | 3350          | 2000                | 1000                 | 58   |
| Tea                     | Zesta          | 1800          | 100                 | 1000                 | 190  |
| Milk powder             | Anchor         | 675           | 100                 | 1000                 | 116  |
|                         | Pelwatta       | 588           | 100                 | 1000                 | 108  |
| Milk                    | Ambewela       | 720           | 2000                | 1000                 | 27   |
| Toffee                  |                | 60000         | 100                 | 1000                 | 1095 |
| Coconut                 |                | 3000          | 100                 | 1000                 | 245  |
| Brooms                  |                | 240           | 100                 | 1000                 | 69   |
| Baked biscuits          |                | 360           | 100                 | 1000                 | 85   |

Table 10: Calculated economic order quantities using proposed excel.

However, for further increased efficiency and to utilize certain availabilities, a thorough analysis of P and Q models were done to understand the most efficient middle-ground. Each item in the Bill of Materials was allocated to either P or Q model accordingly.

### **2.3.3 Model Allocation**

#### **P Model:**

Categorization was based on two reasons,

- Items that have no major contribution to overall sales, which are available as a mean of expanding product variety – Perfume, Party items, Slippers.
  - Product demand is uncertain.
  - Not worth the effort it takes to continuously manage the inventory.
- Unilever products that are supplied through routine vendor visits (once every two weeks) – Washing powder, Liquid wash, Soap, Handwash, Sanitary products, Shampoo.
  - Consistent and routine vendor visits are available.
  - Zero lead time, ordering cost and delivery cost.
  - Guarantees better supplier relationships and discounts.

#### **Q Model:**

A Q model was adapted to the remaining items due to the following reasons,

- Demand for each item is normally distributed and known over a period of time.
- Lead time is constant.
- Ordering and setup costs are constant.

Further categorization based on variability in demand,

- Items with variable demand – Safety stock was determined based on the standard deviation of sales and targeted service probability (Probabilistic approach).
- Items with no variability in demand – A fixed safety stock was set to be maintained as a buffer (Simple deterministic approach).

We identified two major concerns regarding the EOQ of some items and opted for a modified Order Quantity (OQ) for each of those items,

- Average number of days to exhaust EOQ > Acceptable shelf life of perishable items  
⇒  $OQ = \text{Average daily demand} \times \text{Shelf life in days}$
- $EOQ > \text{Maximum storage capacity allocated for the item}$   
⇒  $OQ = \text{Maximum storage capacity allocated for the item}$

| Category                | Item           | Annual Demand | EOQ  | d      | Days to exhaust EOQ | Acceptable Shelf life (Days) | Storage Capacity (Units) | Model |
|-------------------------|----------------|---------------|------|--------|---------------------|------------------------------|--------------------------|-------|
| Ice cream               | Elephant house | 1050          | 32   | 3.50   | 9                   | 60                           | 35                       | Q     |
|                         | Cargills       | 518           | 23   | 1.73   | 13                  | 60                           | 25                       | Q     |
| Perfume                 | Men's          | 120           | -    | 0.40   | -                   | -                            | -                        | P     |
|                         | Women's        | 120           | -    | 0.40   | -                   | -                            | -                        | P     |
| Chicken                 | Bairaha        | 270           | 16   | 0.90   | 18                  | 30                           | 12                       | C     |
| Sausage                 |                | 400           | 20   | 1.33   | 15                  | 45                           | 20                       | Q     |
| Party items             |                | 60            | -    | 0.20   | -                   | -                            | -                        | P     |
| Stationary              | Atlas CR       | 480           | 98   | 1.60   | 61                  | 720                          | 150                      | Q     |
| Daily medicine          | Siddalepa      | 180           | 60   | 0.60   | 100                 | 720                          | 100                      | Q     |
| Toilet paper            | Flora          | 300           | 77   | 1.00   | 77                  | 720                          | 80                       | Q     |
| Incense sticks          | Amritha        | 600           | 110  | 2.00   | 55                  | 720                          | 200                      | Q     |
| Bulbs                   | Phillips       | 72            | 38   | 0.24   | 158                 | 720                          | 60                       | Q     |
| Wicks                   |                | 600           | 110  | 2.00   | 55                  | 720                          | 150                      | Q     |
| Washing powder          | Sunlight       | 1200          | 155  | 4.00   | 39                  | 720                          | 60                       | P     |
| Washing items           | Dettol         | 180           | 60   | 0.60   | 100                 | 720                          | 100                      | Q     |
| Liquid wash             | Vim            | 720           | 120  | 2.40   | 50                  | 720                          | 35                       | P     |
| Mosquito repellent Coil | Ninja          | 120           | 49   | 0.40   | 123                 | 720                          | 100                      | Q     |
|                         | Mortein        | 360           | 85   | 1.20   | 71                  | 720                          | 100                      | Q     |
| Toothpaste              | Signal         | 720           | 120  | 2.40   | 50                  | 720                          | 150                      | Q     |
|                         | Clogard        | 180           | 60   | 0.60   | 100                 | 720                          | 150                      | Q     |
| Salmon                  | Jack mackarel  | 120           | 49   | 0.40   | 123                 | 720                          | 60                       | Q     |
| Soya                    | Raigam         | 300           | 77   | 1.00   | 77                  | 180                          | 100                      | Q     |
|                         | Lankasoy       | 300           | 77   | 1.00   | 77                  | 180                          | 100                      | Q     |
| Potatoes                |                | 360           | 38   | 1.20   | 32                  | 12                           | 25                       | C     |
| Garlic                  |                | 120           | 22   | 0.40   | 55                  | 90                           | 25                       | Q     |
| Onions                  |                | 720           | 54   | 2.40   | 23                  | 75                           | 25                       | C     |
| Pasta                   | Harischandra   | 120           | 49   | 0.40   | 123                 | 180                          | 50                       | Q     |
| Noodles                 | Kottu mee      | 1920          | 196  | 6.40   | 31                  | 180                          | 200                      | Q     |
|                         | Harischandra   | 240           | 69   | 0.80   | 86                  | 180                          | 100                      | Q     |
| Lunch sheet             |                | 720           | 120  | 2.40   | 50                  | 720                          | 150                      | Q     |
| Soap                    | Lux            | 240           | 31   | 0.80   | 39                  | 720                          | 100                      | P     |
| Handwash                | Lifebuoy       | 180           | 60   | 0.60   | 100                 | 720                          | 100                      | P     |
| Envelopes               |                | 360           | 85   | 1.20   | 71                  | 720                          | 100                      | Q     |
| Hair-gel                |                | 60            | 35   | 0.20   | 175                 | 720                          | 50                       | Q     |
| Hair dye                |                | 240           | 69   | 0.80   | 86                  | 720                          | 80                       | Q     |
| Face powder             |                | 36            | 27   | 0.12   | 225                 | 720                          | 35                       | Q     |
| Sanitary products       | Fems           | 900           | 134  | 3.00   | 45                  | 720                          | 50                       | P     |
| Sweets                  | Ritzbury       | 1200          | 49   | 4.00   | 12                  | 180                          | 100                      | Q     |
|                         | Uswatte        | 600           | 35   | 2.00   | 18                  | 180                          | 50                       | Q     |
| Shampoo                 | Sunsilk        | 360           | 85   | 1.20   | 71                  | 720                          | 50                       | P     |
|                         | Dandex         | 120           | 49   | 0.40   | 123                 | 720                          | 50                       | Q     |
| Powdered coconut milk   | Maggie         | 120           | 49   | 0.40   | 123                 | 180                          | 50                       | Q     |
| Soup cubes              | Knorr          | 360           | 85   | 1.20   | 71                  | 180                          | 100                      | Q     |
| Condiments              | Kist           | 180           | 60   | 0.60   | 100                 | 180                          | 80                       | Q     |
| Culinary                | Mas            | 1200          | 155  | 4.00   | 39                  | 180                          | 200                      | Q     |
| Dried fish              |                | 180           | 60   | 0.60   | 100                 | 180                          | 75                       | Q     |
| Eggs                    |                | 1200          | 155  | 4.00   | 39                  | 10                           | 50                       | C     |
| Razor                   | Bic            | 1080          | 147  | 3.60   | 41                  | 720                          | 150                      | Q     |
| Biscuits                | Maliban        | 1440          | 170  | 4.80   | 35                  | 180                          | 200                      | Q     |
|                         | Munchee        | 840           | 130  | 2.80   | 46                  | 180                          | 200                      | Q     |
| Slippers                |                | 120           | 49   | 0.40   | 123                 | 720                          | 50                       | P     |
| Bakery items            | Flour          | 600           | 110  | 2.00   | 55                  | 90                           | 50                       | C     |
|                         | Motha          | 720           | 120  | 2.40   | 50                  | 720                          | 150                      | Q     |
| Soft drinks             | Elephant house | 3350          | 58   | 11.17  | 5                   | 60                           | 100                      | Q     |
| Tea                     | Zesta          | 1800          | 190  | 6.00   | 32                  | 180                          | 200                      | Q     |
| Milk powder             | Anchor         | 675           | 116  | 2.25   | 52                  | 360                          | 50                       | C     |
|                         | Pelwatte       | 588           | 108  | 1.96   | 55                  | 360                          | 50                       | C     |
| Milk                    | Ambewela       | 720           | 27   | 2.40   | 11                  | 14                           | 30                       | Q     |
| Toffee                  |                | 60000         | 1095 | 200.00 | 5                   | 60                           | 2000                     | Q     |
| Coconut                 |                | 3000          | 245  | 10.00  | 25                  | 30                           | 75                       | C     |
| Brooms                  |                | 240           | 69   | 0.80   | 86                  | 720                          | 20                       | C     |
| Baked biscuits          |                | 360           | 85   | 1.20   | 71                  | 90                           | 100                      | Q     |

Table 11: Classification of available items to P and Q models (P – P model, Q – Q model, C – Q model with modified OQ)

| Category                | Item          | Annual Demand | EOQ  | Lead Time (days) | d      | Safety Stock | ROL |
|-------------------------|---------------|---------------|------|------------------|--------|--------------|-----|
| Stationary              | Atlas CR      | 480           | 98   | 3                | 1.60   | 0            | 5   |
| Daily medicine          | Siddalepa     | 180           | 60   | 2                | 0.60   | 1            | 2   |
| Toilet paper            | Flora         | 300           | 77   | 1                | 1.00   | 2            | 3   |
| Incense sticks          | Amritha       | 600           | 110  | 2                | 2.00   | 1            | 5   |
| Bulbs                   | Phillips      | 72            | 38   | 3                | 0.24   | 1            | 2   |
| Wicks                   |               | 600           | 110  | 1                | 2.00   | 3            | 5   |
| Washing items           | Dettol        | 180           | 60   | 2                | 0.60   | 1            | 2   |
| Mosquito repellent Coil | Ninja         | 120           | 49   | 2                | 0.40   | 2            | 3   |
|                         | Mortein       | 360           | 85   | 2                | 1.20   | 0            | 2   |
| Toothpaste              | Signal        | 720           | 120  | 1                | 2.40   | 3            | 5   |
|                         | Clogard       | 180           | 60   | 1                | 0.60   | 2            | 3   |
| Salmon                  | Jack mackarel | 120           | 49   | 1                | 0.40   | 2            | 2   |
| Soya                    | Raigam        | 300           | 77   | 1                | 1.00   | 2            | 3   |
|                         | Lankasoy      | 300           | 77   | 1                | 1.00   | 2            | 3   |
| Garlic                  |               | 120           | 22   | 1                | 0.40   | 2            | 2   |
| Pasta                   | Harischandra  | 120           | 49   | 2                | 0.40   | 1            | 2   |
| Noodles                 | Kottu mee     | 1920          | 196  | 2                | 6.40   | 2            | 15  |
|                         | Harischandra  | 240           | 69   | 2                | 0.80   | 2            | 4   |
| Lunch sheet             |               | 720           | 120  | 2                | 2.40   | 2            | 7   |
| Envelopes               |               | 360           | 85   | 2                | 1.20   | 2            | 4   |
| Hair-gel                |               | 60            | 35   | 3                | 0.20   | 1            | 2   |
| Hair dye                |               | 240           | 69   | 3                | 0.80   | 2            | 4   |
| Face powder             |               | 36            | 27   | 2                | 0.12   | 2            | 2   |
| Sweets                  | Ritzbury      | 1200          | 49   | 1                | 4.00   | 2            | 6   |
|                         | Uswatte       | 600           | 35   | 1                | 2.00   | 2            | 4   |
| Shampoo                 | Dandex        | 120           | 49   | 2                | 0.40   | 1            | 2   |
| Powdered coconut milk   | Maggie        | 120           | 49   | 1                | 0.40   | 3            | 3   |
| Soup cubes              | Knorr         | 360           | 85   | 2                | 1.20   | 3            | 5   |
| Condiments              | Kist          | 180           | 60   | 2                | 0.60   | 2            | 3   |
| Culinary                | Mas           | 1200          | 155  | 2                | 4.00   | 2            | 10  |
| Dried fish              |               | 180           | 60   | 1                | 0.60   | 2            | 3   |
| Razor                   | Bic           | 1080          | 147  | 2                | 3.60   | 2            | 9   |
| Biscuits                | Maliban       | 1440          | 170  | 1                | 4.80   | 2            | 7   |
|                         | Munchee       | 840           | 130  | 1                | 2.80   | 2            | 5   |
| Bakery items            | Motha         | 720           | 120  | 2                | 2.40   | 2            | 7   |
| Tea                     | Zesta         | 1800          | 190  | 2                | 6.00   | 2            | 14  |
| Milk                    | Ambewela      | 720           | 27   | 1                | 2.40   | 2            | 4   |
| Toffee                  |               | 60000         | 1095 | 1                | 200.00 | 20           | 220 |
| Baked biscuits          |               | 360           | 85   | 2                | 1.20   | 2            | 4   |

Table 12: Q model with safety stock | Simple deterministic approach

| Category    | Item           | Annual Demand | EOQ | Lead Time (days) | Standard Deviation per day | $\sigma L$ | P   | Z    | Safety Stock | d     | ROL |
|-------------|----------------|---------------|-----|------------------|----------------------------|------------|-----|------|--------------|-------|-----|
| Ice cream   | Elephant house | 1050          | 32  | 2                | 0.29                       | 0.58       | 99% | 2.33 | 1.35         | 3.50  | 8   |
|             | Cargills       | 518           | 23  | 2                | 0.32                       | 0.65       | 90% | 1.28 | 0.83         | 1.73  | 4   |
| Sausage     |                | 400           | 20  | 1                | 0.26                       | 0.26       | 99% | 2.33 | 0.60         | 1.33  | 2   |
| Soft drinks | Elephant house | 3350          | 58  | 1                | 2.51                       | 2.51       | 99% | 2.33 | 5.84         | 11.17 | 17  |

Table 13: Q model with safety stock | Probabilistic approach.

| Category     | Item  | Annual Demand | EOQ | d     | Days to exhaust EOQ | Acceptable Shelf life (Days) | Storage Capacity (Units) | L | Safety Stock | ROL | Modified OQ |
|--------------|-------|---------------|-----|-------|---------------------|------------------------------|--------------------------|---|--------------|-----|-------------|
| Potatoes     |       | 360           | 38  | 1.20  | 32                  | 12                           | 25                       | 1 | 1            | 2   | 14          |
| Onions       |       | 720           | 54  | 2.40  | 23                  | 75                           | 25                       | 1 | 1            | 3   | 25          |
| Eggs         |       | 1200          | 155 | 4.00  | 39                  | 10                           | 50                       | 1 | 4            | 8   | 40          |
| Bakery items | Flour | 600           | 110 | 2.00  | 55                  | 90                           | 50                       | 2 | 2            | 6   | 50          |
| Coconut      |       | 3000          | 245 | 10.00 | 25                  | 30                           | 75                       | 1 | 5            | 15  | 75          |
| Brooms       |       | 240           | 69  | 0.80  | 86                  | 720                          | 20                       | 3 | 1            | 3   | 20          |

Table 14: Q model | Modified OQ | Simple deterministic approach

| Category    | Item    | Annual Demand | EOQ | Storage Capacity | Lead Time (days) | Standard Deviation per day | $\sigma L$ | P   | Z    | Safety Stock | d    | ROL | Modified OQ |
|-------------|---------|---------------|-----|------------------|------------------|----------------------------|------------|-----|------|--------------|------|-----|-------------|
| Chicken     | Bairaha | 270           | 16  | 12               | 1                | 0.19                       | 0.19       | 99% | 2.33 | 0.45         | 0.90 | 1   | 12          |
| Milk powder | Anchor  | 675           | 116 | 50               | 2                | 0.14                       | 0.29       | 90% | 1.28 | 0.37         | 2.25 | 5   | 50          |
|             | Pelwate | 588           | 108 | 50               | 3                | 0.12                       | 0.35       | 99% | 2.33 | 0.81         | 1.96 | 7   | 50          |

Table 15: Q model | Modified OQ | Probabilistic approach

| Category          | Item     | Annual Demand | L | T   | d    | Safety Stock | I  | q  |
|-------------------|----------|---------------|---|-----|------|--------------|----|----|
| Perfume           | Men's    | 120           | 3 | 120 | 0.40 | 1            | 11 | 39 |
|                   | Women's  | 120           | 3 | 120 | 0.40 | 1            | 8  | 42 |
| Party items       |          | 60            | 3 | 120 | 0.20 | 1            | 5  | 21 |
| Washing powder    | Sunlight | 1200          | 0 | 14  | 4.00 | 3            | 6  | 53 |
| Liquid wash       | Vim      | 720           | 0 | 14  | 2.40 | 3            | 5  | 32 |
| Soap              | Lux      | 240           | 0 | 14  | 0.80 | 3            | 3  | 11 |
| Handwash          | Lifebuoy | 180           | 0 | 14  | 0.60 | 3            | 4  | 7  |
| Sanitary products | Fems     | 900           | 0 | 14  | 3.00 | 3            | 3  | 42 |
| Shampoo           | Sunsilk  | 360           | 0 | 14  | 1.20 | 3            | 4  | 16 |
| Slippers          |          | 120           | 3 | 120 | 0.40 | 1            | 14 | 36 |

Table 16: P model

### 2.3.4 Inventory Management

An excel based user interface was provided to the shop owner for proper inventory management.

#### Q model management,

- Sales and Current stock is updated in real time with the use of interactive controls.
- If, Current stock  $\leq$  ROL  
⇒ Item name and Order quantity is highlighted in red  
⇒ Owner is alerted that restocking needs to be done
- For perishable items,  
If, Days since the recent stock  $\geq$  Acceptable shelf life (with a tolerance for action)  
⇒ Item name and Current stock level is highlighted in yellow

| Category       | Item           | Order Quantity | ROL | Initial Stock | Current Stock | Sales |    | Recent Stock Date |
|----------------|----------------|----------------|-----|---------------|---------------|-------|----|-------------------|
| Ice cream      | Elephant house | 32             | 8   | 32            | 7             | 25    | ▲▼ | 11/5/2023         |
|                | Cargills       | 23             | 4   | 23            | 23            | 0     | ▲▼ | 11/5/2023         |
| Chicken        | Bairaha        | 12             | 1   | 16            | 16            | 0     | ▲▼ | 9/29/2023         |
| Sausage        |                | 20             | 2   | 20            | 1             | 19    | ▲▼ | 11/5/2023         |
| Stationary     | Atlas CR       | 98             | 5   | 98            | 98            | 0     | ▲▼ | 11/5/2023         |
| Daily medicine | Siddalepa      | 60             | 2   | 60            | 60            | 0     | ▲▼ | 11/5/2023         |
| Toilet paper   | Flora          | 77             | 3   | 77            | 77            | 0     | ▲▼ | 11/5/2023         |

*Figure 18: Inventory Management Interface for Q model items*

#### P model management,

- Sales, Inventory at hand and Order quantity is updated in real time with the use of interactive controls.
- Stock date is recorded, and the Item name and Order quantity is highlighted in red when the Review point is reached.

| Category          | Item     | Order Quantity | Initial Stock | Inventory at hand | Sales |        | Stock Date |
|-------------------|----------|----------------|---------------|-------------------|-------|--------|------------|
| Perfume           | Men's    | 39             | 40            | 11                | 29    | ▲<br>▼ | 7/27/2023  |
|                   | Women's  | 42             | 40            | 8                 | 32    | ▲<br>▼ | 7/27/2023  |
| Party items       |          | 21             | 20            | 5                 | 15    | ▲<br>▼ | 7/27/2023  |
| Washing powder    | Sunlight | 53             | 54            | 6                 | 48    | ▲<br>▼ | 10/15/2023 |
| Liquid wash       | Vim      | 32             | 32            | 5                 | 27    | ▲<br>▼ | 10/15/2023 |
| Soap              | Lux      | 11             | 9             | 3                 | 6     | ▲<br>▼ | 10/15/2023 |
| Handwash          | Lifebuoy | 7              | 6             | 4                 | 2     | ▲<br>▼ | 10/15/2023 |
| Sanitary products | Fems     | 42             | 40            | 3                 | 37    | ▲<br>▼ | 10/15/2023 |
| Shampoo           | Sunsilk  | 16             | 15            | 4                 | 11    | ▲<br>▼ | 10/15/2023 |
| Slippers          |          | 36             | 38            | 14                | 24    | ▲<br>▼ | 7/27/2023  |

Figure 19: Inventory Management Interface for P model items

#### Identification of trends,

- Forecasted data is linked to the item list to indicate the demand trends.
- The indication is based on the gradient of the linear forecast.

| Category    | Item           | Linear forecast gradient |
|-------------|----------------|--------------------------|
| Ice cream   | Elephant house | 0.32                     |
|             | Cargills       | -4.80                    |
| Milk powder | Anchor         | -3.92                    |
|             | Pelwatte       | 8.71                     |

Figure 20: Forecasted demand trends predicted by the created excel sheet (Green – Increasing demand, Red – Decreasing demand).

### 2.3.5 Comparison of performance

- In the existing system,
  - Restocking is done based on intuition.
  - Most of the items are checked and stocked at least monthly.  
⇒ Unnecessary and inefficient
  - Order quantity at each restocking is estimated based on experience.  
⇒ Increased possibility of high order frequency
- A significant reduction in Average number of orders per year can be observed after the implemented solution.
- This reduces the ordering and delivery costs associated.

| Category                | Item           | Annual Demand | EOQ | Average number of orders per year |       | Reduction |
|-------------------------|----------------|---------------|-----|-----------------------------------|-------|-----------|
|                         |                |               |     | Before                            | After |           |
| Ice cream               | Elephant house | 1050          | 32  | 36                                | 33    | 3         |
|                         | Cargills       | 518           | 23  | 36                                | 23    | 13        |
| Chicken                 | Bairaha        | 270           | 16  | 24                                | 17    | 7         |
| Sausage                 |                | 400           | 20  | 24                                | 20    | 4         |
| Stationary              | Atlas CR       | 480           | 98  | 6                                 | 5     | 1         |
| Daily medicine          | Siddalepa      | 180           | 60  | 6                                 | 3     | 3         |
| Toilet paper            | Flora          | 300           | 77  | 12                                | 4     | 8         |
| Incense sticks          | Amritha        | 600           | 110 | 6                                 | 5     | 1         |
| Bulbs                   | Phillips       | 72            | 38  | 12                                | 2     | 10        |
| Wicks                   |                | 600           | 110 | 12                                | 5     | 7         |
| Washing items           | Dettol         | 180           | 60  | 12                                | 3     | 9         |
| Mosquito repellent Coil | Ninja          | 120           | 49  | 12                                | 2     | 10        |
|                         | Mortein        | 360           | 85  | 12                                | 4     | 8         |
| Toothpaste              | Signal         | 720           | 120 | 12                                | 6     | 6         |
|                         | Clogard        | 180           | 60  | 12                                | 3     | 9         |
| Salmon                  | Jack mackarel  | 120           | 49  | 12                                | 2     | 10        |
| Soya                    | Raigam         | 300           | 77  | 12                                | 4     | 8         |
|                         | Lankasoy       | 300           | 77  | 12                                | 4     | 8         |
| Potatoes                |                | 360           | 38  | 12                                | 9     | 3         |
| Garlic                  |                | 120           | 22  | 12                                | 5     | 7         |
| Onions                  |                | 720           | 54  | 18                                | 13    | 5         |
| Pasta                   | Harischandra   | 120           | 49  | 12                                | 2     | 10        |
| Noodles                 | Kottu mee      | 1920          | 196 | 12                                | 10    | 2         |
|                         | Harischandra   | 240           | 69  | 12                                | 3     | 9         |
| Lunch sheet             |                | 720           | 120 | 6                                 | 6     | 0         |
| Envelopes               |                | 360           | 85  | 6                                 | 4     | 2         |
| Hair-gel                |                | 60            | 35  | 6                                 | 2     | 4         |
| Hair dye                |                | 240           | 69  | 6                                 | 3     | 3         |
| Face powder             |                | 36            | 27  | 6                                 | 1     | 5         |
| Sweets                  | Ritzbury       | 1200          | 49  | 48                                | 24    | 24        |
|                         | Uswatte        | 600           | 35  | 36                                | 17    | 19        |
| Shampoo                 | Dandex         | 120           | 49  | 12                                | 2     | 10        |
| Powdered coconut milk   | Maggie         | 120           | 49  | 12                                | 2     | 10        |
| Soup cubes              | Knorr          | 360           | 85  | 12                                | 4     | 8         |
| Condiments              | Kist           | 180           | 60  | 12                                | 3     | 9         |
| Culinary                | Mas            | 1200          | 155 | 12                                | 8     | 4         |
| Dried fish              |                | 180           | 60  | 6                                 | 3     | 3         |
| Eggs                    |                | 1200          | 155 | 24                                | 8     | 16        |
| Razor                   | Bic            | 1080          | 147 | 12                                | 7     | 5         |
| Biscuits                | Maliban        | 1440          | 170 | 12                                | 8     | 4         |
|                         | Munchee        | 840           | 130 | 12                                | 6     | 6         |

|                |                |       |      |    |    |    |
|----------------|----------------|-------|------|----|----|----|
| Bakery items   | Flour          | 600   | 110  | 12 | 5  | 7  |
|                | Motha          | 720   | 120  | 12 | 6  | 6  |
| Soft drinks    | Elephant house | 3350  | 58   | 60 | 58 | 2  |
| Tea            | Zesta          | 1800  | 190  | 12 | 9  | 3  |
| Milk powder    | Anchor         | 675   | 116  | 24 | 6  | 18 |
|                | Pelwatta       | 588   | 108  | 24 | 5  | 19 |
| Milk           | Ambewela       | 720   | 27   | 48 | 27 | 21 |
| Toffee         |                | 60000 | 1095 | 72 | 55 | 17 |
| Coconut        |                | 3000  | 245  | 36 | 12 | 24 |
| Brooms         |                | 240   | 69   | 6  | 3  | 3  |
| Baked biscuits |                | 360   | 85   | 12 | 4  | 8  |

*Table 17: Observed reduction in average number of orders placed per year.*

Following are the reasons that guarantees improved performance than the existing system,

- Maintaining accurate sales records enables the use of a probabilistic approach for determining safety stocks for each item.
- Effective demand forecasting is achievable, and this data integration enhances inventory management.
- Forecasted demand can be used in Economic Order Quantity (EOQ) calculations, improving inventory control.
- Forecasted trends inform decisions about service probabilities for safety stock levels.
- These trends guide strategic choices for effectively managing items in the market.
- Proactive monitoring of perishable items minimizes waste through actions like timely sales promotions as shelf life limits approach.

## 2.4 Quality Management [8]

Since this is rather a service than selling their own produced goods, the dimensions of quality which applies to a service is related in this case. They are,

- Time and Timeliness – For this, the customers should spend a minimum time to buy goods and billing process.
- Completeness – Stock availability of goods based on customer preferences.
- Courtesy – The way that the customers are treated by the employees. Customers should be treated with a friendly manner, and they should be given the impression such that they come back to buy different needs.
- Consistency – All the customers should be treated with the same manner.
- Accessibility and convenience – Long ques should not be there which keeps customers waiting for long to obtain the service. Also, there should be a minimum rush for the customers to shop goods at their own pace.
- Accuracy – There should be minimum errors happened in processes like billing and delivery services.

The layout of the facility plays a key role in the aspect of quality control in this “pandula enterprises”. Hence the layout should be planned in such a way that the customers have a minimal waiting time in the billing process and as well as minimum rush through the area to collect goods at their own pace. For this purpose, it is good to have a self-checkout counter for managing the rush on peak hours as well as it is easier for customers who buys small number of items to make a self-checkout.

It was observed from the data that the mostly sold items during the peak hours are daily household items and food items such as ice creams, milk packets in multiple times with small quantities, hence it will be ideal to have a self-checkout counter for these items to avoid formation of long ques.

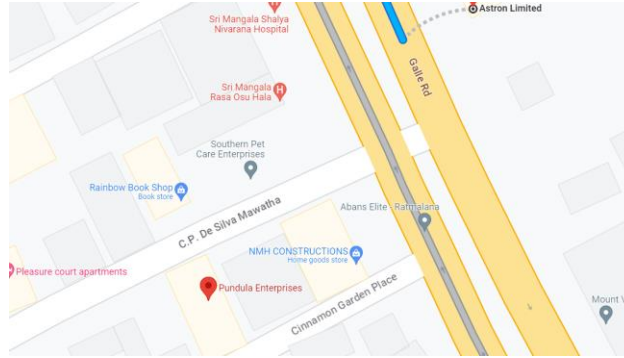
Also, the layout of the supermarket should be arranged in a way such that the amount of time spent by customers in search of their needed goods is minimal. Arranging this for every customer is difficult. Therefore, by analysing the data, we can get the list of items which have maximum daily sales, which will eventually be the most searched items by the customers. Therefore, those items should be located at easily accessible locations where customers do not have to go far away from the billing counter to get them.

Further, to ensure the quality of the goods supplied to the customers, it is important to practice good hygienic practices (GHP) in the supermarket and warehouse by the employees. Some of the key aspects of it are mentioned below.

- **Storage & Handling** – Proper storing of perishable items such as meat, meat products and milk products are important. The proper temperatures and humidity levels should be maintained during this. Also, these items should be prevented from contaminations and spoilage.
- **Cleaning & Sanitation** – Regular cleaning and sanitation is important for preventing the stocks from growth of harmful microorganisms and spread of pathogens.
- **Employee Hygiene** – Employees handling the items should have to be under strict hygiene practices.
- **Pest Control** – Necessary pest control measures should be taken in order to prevent the stocks. Also, it is important that the pest control measures taken should not cause contaminations or any damage to the stocks.
- **Product Inspection** – The available stocks should be checked regularly for damages and expiration. For this, a computerized record keeping is helpful. If there is an issue with a certain product, the records will help to trace the origin of the product and identifying the similar batch of the products and addressing the issue as soon as possible.

## 2.5 Marketing Enhancements [9]

- Currently, the only marketing available is the sign board in front of the entrance. Since this store is situated in a by road near the main road, only the people who lives in the neighbourhood gets to see that. Hence the effectiveness of it is very low. If a sign board can be placed at the entrance to the by road, it can attract the people travelling on the main road as well.
- Also, to increase the market sales, we suggest introducing stock clearance sales and discounts for items which are not being sold frequently but available with ample amounts in the stock.



*Figure 21: Location of the Convenience store.*

### **3 Future Work**

- Introduction of a Self-checkout counter – It was observed from the data that the mostly sold items during the peak hours are daily household items and food items such as ice creams, milk packets in multiple times with small quantities, hence it will be ideal to have a self-checkout counter for these items in order to avoid formation of long ques.
- Extension of Online Delivery Services – Currently, the pick Me and Uber eats services are available for sun crush (Beverages) and cigarettes. This can be extended to other items as well which will increase the sales.
- Premises Extension – Depending on the demand and customer attraction, the premises can be extended up to the nearest building, which is owned by the same owner, which will be beneficial in organizing and upgrading convenience of customers.

## 4 Conclusions

Finally, our group's case study of "Pandula Enterprises," a convenience store in Ratmalana, revealed various management obstacles and potential for development. We discovered critical shortcomings such as a disorganized shelf system, inefficient storage use, a lack of effective invoicing and inventory management systems, and an inability to accurately estimate demand.

Our study sought to address these issues and improve the store's operations. We used a variety of methods, including:

Forecasting Demand using approaches such as exponential smoothing and least square algorithms, we collected sales data and anticipated demand for various items was done. This enabled us to optimize stock levels and order quantities, resulting in increased store profitability.

For planning of facility layout, we modified the store's layout to improve the customer experience and cut wait times. We built three sample layouts with unique features to streamline the shopping process by categorizing products depending on customer preferences.

We implemented inventory models (P and Q models) to better manage stock levels, resulting in a considerable reduction in average annual orders and associated expenses. For the store owner to successfully manage inventory, an interactive Excel interface was created.

Service-related quality dimensions such as time and timeliness, completeness, courtesy, consistency, accessibility, and accuracy were taken into consideration. We advocated for the installation of a self-checkout counter, the arrangement of commonly purchased commodities for convenient access, and the maintenance of appropriate hygiene procedures in the supermarket and warehouse.

Marketing enhancements such as increasing the size of signboards to attract more customers and introducing stock clearance specials and discounts on slow-moving items.

To meet rising demand and improve consumer convenience, we propose implementing a self-checkout counter, expanding online delivery services, and expanding the premises.

In summary, our research not only identified Pandula Enterprises' difficulties, but also gave real solutions to optimize operations, improve the customer experience, and increase profitability.

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## Appendices

[https://drive.google.com/drive/folders/1v1o1PT6VEmksPKKB\\_fTInFws57SS7Qb?usp=sharing](https://drive.google.com/drive/folders/1v1o1PT6VEmksPKKB_fTInFws57SS7Qb?usp=sharing)